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INVARIANT TWO-JETS AND EFFECTIVE HYPERBOLICITY FOR COMPLEMENTS OF TWO PLANE CURVES

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ABSTRACT. Let $D = C_1 + C_2 \subset \mathbb{P}^2$ be a simple-normal-crossing union of smooth plane curves of degrees $1 \leqslant d_1 \leqslant d_2$. We prove an effective Second Main Theorem on a nonempty Zariski-open set of ordered pairs whenever

$$d_1, d_2 \geqslant 3, \qquad \text{or} \qquad d_1 = 2,\ d_2 \geqslant 5, \qquad \text{or} \qquad d_1 = 1,\ d_2 \geqslant 8.$$

After intersecting with the standard very-general algebraic-hyperbolicity locus, we obtain Kobayashi hyperbolicity and hyperbolic embedding of the complement. The proof uses logarithmic invariant two-jet differentials, the corrected Demailly–El Goul zero-locus argument, and a two-component adaptation of slanted vector fields with base twist $\mathcal{O}_{\mathbb{P}^2}(3)$. The adapted twist three is uniform in (d_1, d_2): simultaneous cancellation of the two root equations requires third Taylor differences, but no seventh-order polarization. For a linear component the third-remainder space vanishes, and ten compatible mixed generators replace the generic Taylor package. A numerical phase diagram shows that exact finite computation is needed only for the six boundary pairs $(3,3), (3,4), (2,6), (2,5), (1,9), (1,8)$. Their key vanishings are certified by full-column-rank matrices over finite fields. For two cubics, a strengthened range reduces to twelve theorem-relevant degree–twist cases; two low-weight self-tests bring the computational total to 168 character–eigenvalue blocks. The exact source, certificates, and reproduction instructions form an accompanying computational archive. In the two-cubic case the resulting explicit estimate is

$$T_f(r) \leqslant 57\big(N_f^{[1]}(r, C_1) + N_f^{[1]}(r, C_2)\big) + o(T_f(r)) \ \|.$$

For a conic and a quintic one may take the explicit coefficient 45, and for a line and an octic one may take 69.

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1. Introduction

1.1. **Historical setting.** The logarithmic hyperbolicity problem asks how the geometry of a divisor $D \subset \mathbb{P}^n$ constrains entire curves in $\mathbb{P}^n \setminus D$. In dimension one, the model is Picard's theorem: the complement of three points in $\mathbb{P}^1$ admits no nonconstant entire curve. In higher dimension the guiding principle is that sufficiently positive logarithmic cotangent geometry should force the same rigidity. The Green–Griffiths construction and Demailly's reparametrization-invariant refinement turn this principle into differential equations on jets of entire curves. In the logarithmic setting this leads to the bundles

$$E_{k,m}T^*_X(\log D),$$

whose negatively twisted sections vanish along every entire curve in the complement; see [2, 4]. The central difficulty is not merely to produce one equation by Riemann–Roch, but to control its base locus well enough to obtain an independent equation.

For logarithmic surfaces, two complementary ideas made this strategy effective. Miyaoka's zero-locus method (see also Lu–Yau), in the form used by Demailly–El Goul [15, 12, 3], produces a second equation intrinsically on an irreducible component of the first equation's zero divisor. Siu's slanted vector-field method [20], developed effectively in dimension two by Păun [16] and in the logarithmic setting by Rousseau [17], uses the natural action of invariant fields on the universal jet space to turn the first equation ω_1 into a genuine global second jet

differential ω_2. The gain from differentiation is governed by the pole order of the available vector fields; this is why replacing a twist seven construction by twist three has a direct quantitative consequence for a Second Main Theorem.

For the complement of one very general smooth plane curve, Rousseau obtained the long-standing degree bound $d \geqslant 14$ [17, Theorem 2]. Recent work of Hou–Huynh–Merker–Xie lowered it to $d \geqslant 12$ by combining invariant two-jets with a finite key-vanishing calculation [7]; a subsequent symmetry method sharpened the computational framework further [9]. A related three-component model—three general conics—admits a particularly simple logarithmic frame and yields an effective Second Main Theorem [6]. These developments show that low-degree questions are controlled by a delicate interface between intrinsic jet geometry, explicit slanted fields, and exact finite linear algebra.

The chronological starting point is the three-conics paper [6]. Its decisive innovation is the exponent bound that converts an a priori infinite jet-vanishing problem into finite exact algebra; its Second Main Theorem argument also makes essential use of the geometric structure of the Wronskian. The later one-component paper [7] develops this discovery into a one-component theory, including the local invariant-two-jet normal form and the first concrete implementation of the finite Key Vanishing strategy. The higher-direct-image, zero-locus, and algebraic-extension ingredients are inherited from the preceding work; at each use we identify both the source and the two-component specialization. The symmetry organization and the pole-order-three acceleration package follow [9, Section 2]; Section 5 gives the complete two-component adaptation. The principal new technical result here is the Key Vanishing Lemma for two components: its coupled normal form, unequal-degree divisibility conditions, symmetry decomposition, and exact finite-rank certificates do not occur in any of the three preceding papers. This is the fourth paper of the program, but it is not merely an application of the preceding strategy: the two-component vanishing requires a new coupled proof with two residue directions, degree-sensitive overlap equations, and exact symmetry blocks adapted to ordered pairs. The remaining ingredients require genuine two-component adaptations, although we do not present the overall strategy itself as new.

The case of two components has its own history. Let

$$D = C_1 + C_2 \subset \mathbb{P}^2, \qquad 1 \leqslant d_1 := \deg C_1 \leqslant d_2 := \deg C_2,$$

where C_1 and C_2 are smooth and transverse. For very general ordered pairs, Rousseau's first results [18, Théorèmes 1.3 and 2.14] prove hyperbolicity in the ranges $d_1 \geqslant 5$, $d_1 = 4, d_2 \geqslant 7$, $(d_1, d_2) = (4, 4)$, $d_1 = 3, d_2 \geqslant 9$, and $d_1 = 2, d_2 \geqslant 12$. His later logarithmic vector-field construction [17, Theorem 3], also for very general pairs, improves these to $d_1 \geqslant 4$, $d_1 = 3, d_2 \geqslant 5$, $d_1 = 2, d_2 \geqslant 8$, and $d_1 = 1, d_2 \geqslant 11$. The present paper reaches every pair with $d_1, d_2 \geqslant 3$, every pair with $d_1 = 2, d_2 \geqslant 5$, and every pair with $d_1 = 1, d_2 \geqslant 8$. In particular, it settles the two-cubic and cubic–quartic cases, lowers the degree-two boundary from 8 to 5, and lowers the linear-component boundary from 11 to 8.

Remark 1.1 (The line–septic frontier). For $(d_1, d_2) = (1, e)$ one has

$$\bar{c}_1^2 = (e-2)^2, \qquad \bar{c}_2 = (e-1)^2, \qquad A(1, e) = 4e^2 - 34e + 43.$$

At $e = 10$ this gives

$$A(1, 10) = 103, \qquad \rho_{\mathrm{DEG}}(1, 10) = \frac{103}{96} > 1.$$

Thus no finite Key Vanishing Lemma is needed for $e \geqslant 10$. For $e = 9$ one has $\rho_{\mathrm{DEG}} = 61/84$, leaving only $(m, t) = (3, 3), (4, 3)$; for $e = 8$ one has $\rho_{\mathrm{DEG}} = 3/8$, leaving

$$(3, 2), (4, 2), (5, 2), (6, 3), (7, 3), (8, 3).$$

Lemma 1.2 certifies precisely these eight cases, and the degree-one construction in Section 5 supplies the required pole-three differentiation package. This is why the theorem begins at $e = 8$.

At $e = 7$ one has $A(1,7) = 1$ and $\rho_{\mathrm{DEG}} = 1/60$; the required minimal profile is

t	m
1	$3 \leqslant m \leqslant 60$
2	$61 \leqslant m \leqslant 120$
3	$121 \leqslant m \leqslant 180.$

Since $A(1,6) = -17$, the pair $(1,7)$ is the formal Riemann–Roch endpoint of this two-jet architecture. It should not be confused with a predicted Key Vanishing endpoint: $A(1,7)$ is barely positive, the list contains 178 minimal cases, and the numerics make it plausible that at least one of the corresponding negatively twisted spaces is nonzero. No claim for $(1,7)$ is made here.

The two-cubic case is especially close to the conjectural low-degree boundary. Its divisor has only two irreducible components and total degree six. Thus it has the same total degree as the three-conics model but one fewer component, while total degree six is only one above the usual plane Kobayashi benchmark $2\dim\mathbb{P}^2+1 = 5$. The reducible two-component family is of course not the full parameter space of irreducible plane curves of degree six, so these statements are not formally instances of the same genericity conjecture. Nevertheless, simultaneously lowering the number of components and keeping the total degree this small removes much of the logarithmic positivity available in hyperplane- or many-component configurations. In this precise sense, the two-cubic theorem illustrates the low-degree reach of the invariant 2-jet method.

Throughout, a property holds for a *general* ordered pair if it holds on some nonempty Zariski-open subset of the relevant parameter space. It holds for a *very general* ordered pair if its failure locus is contained in a countable union of proper Zariski-closed subsets. We retain the order even when $d_1 = d_2$; no quotient by component exchange is taken.

1.2. **What must be adapted for two components.** Put

$$s = d_1 + d_2, \qquad \kappa = s - 3,$$

and write

$$\overline{T}^* = T^*_{\mathbb{P}^2}(\log D), \qquad \overline{T} = T_{\mathbb{P}^2}(-\log D), \qquad \overline{K} = K_{\mathbb{P}^2} + D \simeq \mathcal{O}_{\mathbb{P}^2}(\kappa).$$

We also write

$$h = c_1(\mathcal{O}_{\mathbb{P}^2}(1)), \qquad \bar{c}_i = c_i(\overline{T}^*).$$

Unless another space is displayed, $\mathcal{O}(q)$ means $\mathcal{O}_{\mathbb{P}^2}(q)$; for an affine variety U, the notation $\mathcal{O}(U)$ denotes its ring of regular functions. The global Demailly–Semple strategy is inherited from the three works just cited. Its implementation for two components is nevertheless not obtained by applying the one-component formulas twice. We record the necessary adaptations, distinguishing them from the main new key-vanishing argument.

First, the universal root cover has two logarithmic root variables, and the mixed vector fields must satisfy the tangency equations for both components with a common quadratic acceleration in the base variables. We construct these fields by prescribing their jet components and correcting their parameter polynomials by third-order Taylor differences. For $d_1, d_2 \geqslant 2$, this correction gives parameter polynomials of the required degrees without changing their Taylor data through order two. Coefficient normalization and the projective coordinate changes then yield a finite family of global fields with base twist $\mathcal{O}_{\mathbb{P}^2}(3)$. To prove independence of the differentiated equation, we work on a fixed jet fiber. The simultaneous gradient- and Hessian-cancellation equations determine the normal rows and the common quadratic acceleration. Third differences absorb the remaining parameter components when $d_i \geqslant 3$; for $d_i = 2$, the corrected parameter polynomial already vanishes once its Taylor data through order two vanish. When $d_1 = 1$, a compatibility condition replaces the degree correction in the linear block, and ten explicit mixed generators supply the required data. After all non-jet components have been cancelled on the fixed fiber, a commutator calculation proves

independence and hence selects a derivative by one of the original global fields. Combined with the zero-locus estimates and finite vanishing results, this construction gives the stated degree ranges, including $d_1 = 2,\ d_2 \geqslant 5$ and $d_1 = 1,\ d_2 \geqslant 8$.

Second, the numerical problem is genuinely two-dimensional. The logarithmic Chern numbers depend separately on d_1 and d_2, and the competition between one-jets, invariant two-jets, and the zero-locus method produces a phase diagram in the (d_1, d_2)-plane. The one-jet Segre number is

$$\bar{c}_1^2 - \bar{c}_2 = d_1 d_2 - 3(d_1 + d_2) + 6.$$

Thus one-jets already treat $d_1 \geqslant 5$ and $d_1 = 4, d_2 \geqslant 7$, but they do not see the low-degree corner addressed here. Invariant two-jets reduce that corner to six boundary pairs $(3,3), (3,4), (2,6), (2,5), (1,9), (1,8)$.

Third, the key vanishing calculation is coupled. For one component there is one residue direction and one defining polynomial. Here chart changes mix two logarithmic residues, and unequal degrees carry different homogeneities. We derive a finite normal form that records these coupled transition and divisibility conditions as sparse integral matrices. Equal-degree symmetry splits the two-cubic matrices into character and involution blocks; the unequal-degree pairs require separate normal forms and carefully chosen SNC test divisors. Exact full-column rank modulo one good prime proves rank over $\mathbb{Q}$, and semicontinuity then gives vanishing on a parameter open. This is an exact proof certificate, not a numerical floating-point test. The symmetry decomposition and exact modular elimination make the finite computations needed to obtain the coefficient 57 in the two-cubic case feasible on a standard laptop. The supplementary material provides the computational records and the instructions needed for independent verification.

The principal technical originality of the paper is the third point: the coupled finite key-vanishing theorem and its proof certificates. The first point is a nontrivial two-component adaptation of the $\mathcal{O}(3)$ mechanism of [9], and the second is the numerical bookkeeping needed to deploy the inherited strategy in the (d_1, d_2)-plane. Together these ingredients retain pole order three in the two-component setting and make the new degree range possible.

1.3. **The finite Key Vanishing Lemma.** For $d \geqslant 1$ write $V_d = H^0(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(d))$. For fixed degrees put

$$\mathcal{S}_{d_1,d_2} = \mathbb{P}(V_{d_1}) \times \mathbb{P}(V_{d_2}),$$

and denote by $\mathcal{S}^{\mathrm{snc}}_{d_1,d_2}$ its nonempty open simple-normal-crossing locus. Define

$$A(d_1, d_2) = 4(d_1^2 + d_2^2) + 17 d_1 d_2 - 51(d_1 + d_2) + 90,$$
$$\rho_{\mathrm{DEG}}(d_1, d_2) = \frac{A(d_1, d_2)}{12(d_1 + d_2 - 3)}.$$

For a genuine second-level divisor only $m \geqslant 3$ occurs. For the five boundary pairs other than $(3,3)$ set

$$\mathfrak{F}_{\mathrm{bd}}(d_1, d_2) = \left\{ (m, t) \in \mathbb{Z}_{\geqslant 3} \times \{1, 2, 3\} :\ \frac{t}{m} \geqslant \rho_{\mathrm{DEG}}(d_1, d_2) \right\}.$$

For the conic–quintic pair we certify one additional minimal case and write

$$\mathfrak{F}_{45}(2,5) = \left\{ (m, t) \in \mathbb{Z}_{\geqslant 3} \times \{1, 2, 3\} :\ \frac{t}{m} > \frac{1}{3} \right\} = \mathfrak{F}_{\mathrm{bd}}(2,5) \cup \{(8,3)\}.$$

For two cubics we use the stronger range

$$\mathfrak{F}_{57}(3,3) = \{ (m, t) \in \mathbb{Z}_{\geqslant 1} \times \{1, 2, 3\} :\ 1 \leqslant m \leqslant 5t - 1 \}.$$

The two cases with $m = 1, 2$ are included as low-weight consistency tests; the geometric argument uses only the twelve cases with $m \geqslant 3$.

Lemma 1.2 (Finite key vanishing)**.** *For each of the six degree pairs*

$$(d_1,d_2)\in\{(3,3),(3,4),(2,6),(2,5),(1,9),(1,8)\}$$

there exists a nonempty Zariski-open subset $\mathcal{U}_{\mathrm{van}}(d_1,d_2)\subset\mathcal{S}^{\mathrm{snc}}_{d_1,d_2}$ *such that, for every parameter* a *in this subset, one has*

$$H^0\big(\mathbb{P}^2,E_{2,m}T^*_{\mathbb{P}^2}(\log D_a)\otimes\mathcal{O}_{\mathbb{P}^2}(-t)\big)=0 \tag{1.1}$$

for every $(m,t)\in\mathfrak{F}_{57}(3,3)$ *when* $(d_1,d_2)=(3,3)$*, for every* $(m,t)\in\mathfrak{F}_{45}(2,5)$ *when* $(d_1,d_2)=(2,5)$*, and for every* $(m,t)\in\mathfrak{F}_{\mathrm{bd}}(d_1,d_2)$ *for the other four pairs.*

The finite sets, after reduction to the least twist for each m, are shown below. The second column is the zero-locus threshold $\rho_{\mathrm{DEG}}(d_1,d_2)$, and the third column lists the pairs (m,t) for which the vanishing (1.1) must be checked. Thus the cubic row records fourteen computations: the four pairs $(m,1)$ with $1\leqslant m\leqslant 4$, the five pairs $(m,2)$ with $5\leqslant m\leqslant 9$, and the five pairs $(m,3)$ with $10\leqslant m\leqslant 14$; its first two entries are the self-tests just described.

(d_1,d_2)	ρ_{DEG}	minimal pairs (m,t)
$(3,3)$	$1/4$	$1\leqslant m\leqslant 4$ for $t=1$; $5\leqslant m\leqslant 9$ for $t=2$; $10\leqslant m\leqslant 14$ for $t=3$
$(3,4)$	$37/48$	$(3,3)$
$(2,6)$	$23/30$	$(3,3)$
$(2,5)$	$19/48$	$(3,2),(4,2),(5,2),(6,3),(7,3),(8,3)$
$(1,9)$	$61/84$	$(3,3),(4,3)$
$(1,8)$	$3/8$	$(3,2),(4,2),(5,2),(6,3),(7,3),(8,3)$

Multiplication by a nonzero homogeneous polynomial handles all larger twists with the same m. Exact modular full rank at one SNC pair implies rank over $\mathbb{Q}$, hence over $\mathbb{C}$; upper semicontinuity then promotes the vanishing to the asserted parameter open. The computational archive contains the case list, matrix dimensions, ranks, nullities, and the data needed for independent verification.[1]

1.4. **Main statements.**

Theorem 1.3 (Effective Second Main Theorem)**.** *Let* $1\leqslant d_1\leqslant d_2$ *be integers. Assume that*

$$d_1,d_2\geqslant 3,\qquad \text{or}\qquad d_1=2,\ d_2\geqslant 5,\qquad \text{or}\qquad d_1=1,\ d_2\geqslant 8.$$

There exist a nonempty Zariski-open subset $\mathcal{U}_{\mathrm{SMT}}(d_1,d_2)\subset\mathcal{S}^{\mathrm{snc}}_{d_1,d_2}$ *and an effectively computable constant* $\mathcal{A}_{d_1,d_2}>0$*, both depending only on the ordered degree pair. For every* a *in this subset, write* $D_a=C_{1,a}+C_{2,a}$*. Then every algebraically nondegenerate entire curve* $f:\mathbb{C}\to\mathbb{P}^2$ *satisfies*

$$T_f(r)\leqslant\mathcal{A}_{d_1,d_2}\,N_f^{[1]}(r,D_a)+o(T_f(r))\ \|, \tag{1.2}$$

where, throughout the paper, truncation for a reduced divisor is taken componentwise:

$$N_f^{[1]}(r,D_a)=N_f^{[1]}(r,C_{1,a})+N_f^{[1]}(r,C_{2,a}).$$

For $(d_1,d_2)=(3,3)$ *one may take* $\mathcal{A}_{3,3}=57$*; for* $(d_1,d_2)=(2,5)$ *one may take* $\mathcal{A}_{2,5}=45$*; and for* $(d_1,d_2)=(1,8)$ *one may take* $\mathcal{A}_{1,8}=69$*.*

Let $\mathcal{S}_{\mathrm{Chen}}(d_1,d_2)$ denote the complement of the countable union of proper closed subsets excluded by Theorem 2.13.

Corollary 1.4 (Hyperbolicity and hyperbolic embedding)**.** *Under the degree assumptions of Theorem 1.3, the complement* $\mathbb{P}^2\setminus D_a$ *is Kobayashi hyperbolic and hyperbolically embedded in* $\mathbb{P}^2$ *for every* $a\in\mathcal{U}_{\mathrm{SMT}}(d_1,d_2)\cap\mathcal{S}_{\mathrm{Chen}}(d_1,d_2)$*. In particular, this conclusion holds for a very general ordered pair of smooth transverse curves in the stated degree range.*

[1]The computational archive is available at https://xiesongyan.github.io/codes/hou_wang_xie_two_plane_curves_cpp_evidence_2026.zip.

Remark 1.5 (Genericity conditions and a subsequent strengthening). Xie–Zhao have strengthened the hyperbolicity and hyperbolic-embedding conclusions of Corollary 1.4 from very general to general ordered pairs in the same degree ranges [22, Theorem F].

In our proof, the two genericity conditions arise at different stages. The effective Second Main Theorem holds on the nonempty Zariski-open subset $\mathcal{U}_{\mathrm{SMT}}$, whereas the application of Xi Chen's theorem requires the very-general locus

$$\mathcal{S}_{\mathrm{Chen}} = \mathcal{S}^{\mathrm{snc}}_{d_1,d_2} \setminus \bigcup_{\nu\geqslant 1} Z_\nu,$$

where each Z_ν is a proper closed subset. Since $\mathcal{S}^{\mathrm{snc}}_{d_1,d_2}$ is irreducible and $\mathcal{U}_{\mathrm{SMT}}$ is nonempty and open, its complement is also a proper closed subset. Consequently,

$$\mathcal{U}_{\mathrm{SMT}} \cap \mathcal{S}_{\mathrm{Chen}} = \mathcal{S}^{\mathrm{snc}}_{d_1,d_2} \setminus \left(\left(\mathcal{S}^{\mathrm{snc}}_{d_1,d_2} \setminus \mathcal{U}_{\mathrm{SMT}}\right) \cup \bigcup_{\nu\geqslant 1} Z_\nu \right)$$

is a nonempty very-general subset over $\mathbb{C}$. Thus our effective estimate holds on a Zariski-open locus, while our deduction of hyperbolicity also uses a very-general condition to exclude algebraically degenerate entire curves.

1.5. **Proof strategy.** Riemann–Roch and logarithmic H^2 vanishing first produce a negatively twisted invariant two-jet differential. The Demailly–El Goul reduction then selects an irreducible factor ω_1 of weighted degree m and twist $-t$, without decreasing the twist-to-weight ratio. If $f^*\omega_1 \not\equiv 0$, the jet Second Main Theorem gives the estimate directly. A zero component pulled back from the first Semple level is handled by McQuillan's theorem. For a genuine second-level component, $t > 3$ is handled by differentiating with the mixed $\mathcal{O}(3)$ fields. If $t \leqslant 3$, the numerical threshold either places t/m in the zero-locus range or reduces the problem to the finite set in Lemma 1.2; the restricted-bigness argument then supplies a second section on the zero divisor. Finally, Xi Chen's inequality excludes algebraically degenerate entire curves from a very general complement.

The explicit endpoint constants use sharper versions of this dichotomy. For a conic and a quintic, the additional certified case $(m,t) = (8,3)$ moves the split to $\rho = t/m = 1/3$. The branch $\rho \leqslant 1/3$ has zero-locus coefficient strictly smaller than 45, whereas $\rho > 1/3$ forces $t \geqslant 4$ and $m \leqslant 3t-1$, giving differentiated coefficient at most 11. Choosing the initial ratio above $1/10$ bounds the direct coefficient strictly below 10.

For two cubics the strengthened finite vanishing range moves the split to $\rho = 1/5$. When $\rho \leqslant 1/5$, the limiting zero-locus coefficient is

$$\frac{57+\sqrt{3189}}{2} < 57.$$

Thus the coefficient in the zero-locus estimate can be chosen below 57. When $\rho > 1/5$, the finite theorem excludes $t \leqslant 3$; hence $t \geqslant 4$, and integrality gives $m \leqslant 5t-1$. The $\mathcal{O}(3)$ derivative therefore gives

$$\frac{m}{t-3} \leqslant \frac{5t-1}{t-3} \leqslant 19.$$

Together with the direct and McQuillan branches, this proves the stated integer constant 57.

Section 9 explains why the stated coefficients suffice to demonstrate the method.

Section 2 collects the logarithmic-jet notation and external analytic inputs. Section 3 computes the logarithmic Chern numbers and the two numerical thresholds. Section 4 constructs the first differential. Section 5 proves the simultaneous mixed $\mathcal{O}(3)$ construction, which is the geometric core of the paper. Section 6 combines the zero-locus and differentiation mechanisms, and Section 7 assembles the main results. Section 8 proves the finite Key Vanishing Lemma by symmetry-adapted exact linear algebra; the supporting computations and verification instructions are provided in the supplementary material. The final Section 9 discusses the role of the explicit constants.

2. Logarithmic jets and external inputs

2.1. **Nevanlinna functions.** Let ω_{FS} be the Fubini–Study form. We use the standard Nevanlinna functions and normalization; see [19]. For $f : \mathbb{C} \to \mathbb{P}^2$, set

$$T_f(r) = \int_1^r \frac{ds}{s} \int_{|z|<s} f^*\omega_{\mathrm{FS}}.$$

If $E \subset \mathbb{P}^2$ is an irreducible divisor and $f(\mathbb{C}) \not\subset E$, let

$$N_f^{[1]}(r, E) = \int_1^r \sum_{|z|<s} \min\{1, \operatorname{ord}_z f^*E\} \frac{ds}{s}.$$

For the reduced divisor $D = C_1 + C_2$, we use the componentwise convention

$$N_f^{[1]}(r, D) := N_f^{[1]}(r, C_1) + N_f^{[1]}(r, C_2).$$

The symbol $\|$ means that the inequality holds for all $r > 1$ outside a set of finite Lebesgue measure. For an algebraically nondegenerate curve in $\mathbb{P}^2$, the standard growth lemma gives

$$O(\log T_f(r) + \log r) = o(T_f(r)) \ \|,$$

so we shall replace the logarithmic error term by $o(T_f)$ after applying a jet-differential estimate.

2.2. **Invariant logarithmic two-jets.** We follow the standard invariant logarithmic jet notation of [2, 4]. Locally on a logarithmic surface, an invariant two-jet differential of weighted degree m has the form

$$\sum_{0\leqslant j\leqslant \lfloor m/3\rfloor} \sum_{\alpha_1+\alpha_2=m-3j} R_{\alpha_1,\alpha_2,j} (f_1')^{\alpha_1} (f_2')^{\alpha_2} (f_1'f_2'' - f_2'f_1'')^j. \tag{2.1}$$

Its associated graded bundle is

$$\mathrm{Gr}^\bullet E_{2,m}\overline{T}^* = \bigoplus_{0\leqslant j\leqslant \lfloor m/3\rfloor} S^{m-3j}\overline{T}^* \otimes \overline{K}^j. \tag{2.2}$$

2.3. **The logarithmic Demailly–Semple tower.** Set $X_0 = \mathbb{P}^2$ and $V_0 = \overline{T}$. We use the convention that $\mathbb{P}(V)$ parametrizes lines in V, so that $\mathcal{O}_{\mathbb{P}(V)}(-1)$ is the tautological subbundle. Let

$$X_1 = \mathbb{P}(V_0), \qquad X_2 = \mathbb{P}(V_1),$$

be the first two levels of the logarithmic Demailly–Semple tower, with projections $\pi_{2,1} : X_2 \to X_1$ and $\pi_{2,0} : X_2 \to X_0$. Write

$$u_1 = c_1\big(\pi_{2,1}^*\mathcal{O}_{X_1}(1)\big), \qquad u_2 = c_1\big(\mathcal{O}_{X_2}(1)\big), \qquad h = \pi_{2,0}^* c_1\big(\mathcal{O}_{\mathbb{P}^2}(1)\big),$$

and

$$\mathcal{O}_{X_2}(a, b) := \pi_{2,1}^*\mathcal{O}_{X_1}(a) \otimes \mathcal{O}_{X_2}(b).$$

Let $\Gamma_2 \subset X_2$ be the vertical divisor at the second stage. With the above convention,

$$\mathcal{O}_{X_2}(\Gamma_2) \simeq \mathcal{O}_{X_2}(-1, 1), \qquad \mathcal{O}_{X_2}(3) \otimes \mathcal{O}_{X_2}(-2\Gamma_2) \simeq \mathcal{O}_{X_2}(2, 1).$$

Let $\mathcal{O}_{X_2}(m)$ denote the standard level-two Demailly–Semple tautological bundle of weighted degree m. The direct-image formula identifies invariant two-jets with its sections:

$$(\pi_{2,0})_*\mathcal{O}_{X_2}(m) \simeq E_{2,m}\overline{T}^*. \tag{2.3}$$

We shall also use the following more precise direct-image statement for the bundle $\mathcal{O}_{X_2}(2, 1)$. Its two-stage filtration proof is the one used in [6, Proposition 6.1]. Keeping every negative-degree graded piece gives the sharp criterion in terms of the single defect $a_1 - 2a_2$; we record the resulting statement and proof for later use.

Proposition 2.1 (Sharp higher direct images)**.** *Let $a_1, a_2 \in \mathbb{Z}$. If $a_2 = -1$, then*

$$R^q(\pi_{2,0})_*\mathcal{O}_{X_2}(a_1, -1) = 0 \qquad (q \geqslant 0).$$

Suppose that $a_2 \geqslant 0$ and put $d = a_1 - 2a_2$. Then

$$R^q(\pi_{2,0})_*\mathcal{O}_{X_2}(a_1, a_2) = 0 \qquad (q \geqslant 2),$$

and

$$R^1(\pi_{2,0})_*\mathcal{O}_{X_2}(a_1, a_2) = 0 \quad \Longleftrightarrow \quad d \geqslant -1.$$

If $d \leqslant -2$, the nonzero R^1 has a finite filtration with successive quotients

$$\overline{K}^{a_2-k-1} \otimes \mathrm{Sym}^{-d-3k-2}\,\overline{T}, \qquad 0 \leqslant k \leqslant \min\left\{a_2, \left\lfloor \frac{-d-2}{3} \right\rfloor\right\}. \tag{2.4}$$

Proof. Write $q = \pi_{2,1}$ and $\pi = \pi_{1,0}$. If $a_2 = -1$, the restriction to each q-fiber is $\mathcal{O}_{\mathbb{P}^1}(-1)$, so every direct image under q, and hence under $\pi_{2,0}$, vanishes. Assume $a_2 \geqslant 0$. The projective-bundle formula gives

$$R^j q_*\mathcal{O}_{X_2}(a_1, a_2) = 0 \quad (j > 0), \qquad q_*\mathcal{O}_{X_2}(a_1, a_2) = \mathcal{O}_{X_1}(a_1) \otimes \mathrm{Sym}^{a_2} V_1^*.$$

The dual directed sequence

$$0 \longrightarrow \mathcal{O}_{X_1}(1) \longrightarrow V_1^* \longrightarrow \pi^*\overline{K} \otimes \mathcal{O}_{X_1}(-2) \longrightarrow 0$$

induces a filtration whose graded quotients are

$$\pi^*\overline{K}^{a_2-k} \otimes \mathcal{O}_{X_1}(d + 3k), \qquad 0 \leqslant k \leqslant a_2. \tag{2.5}$$

For the rank-two projective bundle of lines, $R^1\pi_*\mathcal{O}_{X_1}(n) = 0$ for $n \geqslant -1$, while relative Serre duality gives, for $n \leqslant -2$,

$$R^1\pi_*\mathcal{O}_{X_1}(n) \simeq \mathrm{Sym}^{-n-2}\,\overline{T} \otimes \det \overline{T} = \mathrm{Sym}^{-n-2}\,\overline{T} \otimes \overline{K}^{-1}.$$

Applying this to (2.5) gives precisely the quotients (2.4). In the bad range $d + 3k \leqslant -2$, the corresponding ordinary direct image is zero; ordering the filtration from the good range to the bad range shows through the long exact sequences that these R^1 terms assemble by successive extensions. There are no direct images in degree at least two because both stages have relative dimension one and the first push has no positive higher direct images. The vanishing criterion is therefore exactly $d \geqslant -1$. □

Remark 2.2 (Fiberwise interpretation). On a fiber $F \simeq \mathbb{F}_3$ of $\pi_{2,0}$,

$$\mathcal{O}_{X_2}(a_1, a_2)|_F \simeq \mathcal{O}_{\mathbb{F}_3}\big(a_2E + (a_1 + a_2)f\big),$$

and for $a_2 \geqslant 0$ its cohomology is controlled by the line-bundle degrees $d + 3k$, $0 \leqslant k \leqslant a_2$. Thus the defect $a_1 - 2a_2$ is sharp; conditions involving a_1 and a_2 separately do not suffice.

Lemma 2.3 (Direct images of $\mathcal{O}_{X_2}(2p, p)$)**.** *For every integer $p \geqslant 1$ one has*

$$(\pi_{2,0})_*\mathcal{O}_{X_2}(2p, p) \simeq E_{2,3p}\overline{T}^*. \tag{2.6}$$

Moreover,

$$R^q(\pi_{2,0})_*\mathcal{O}_{X_2}(2p, p) = 0 \qquad (q \geqslant 1). \tag{2.7}$$

Proof. The identification (2.6) is the standard Demailly–Semple direct-image formula for the weight-$3p$ subbundle; see [3, Lemma 3.3(c)]. The higher direct images vanish by Proposition 2.1, because $a_2 = p \geqslant 0$ and $a_1 - 2a_2 = 2p - 2p = 0$. □

Thus the p-th power of $\mathcal{O}_{X_2}(2, 1)$ corresponds canonically to weighted degree $3p$.

We shall repeatedly use the following elementary fact about lifted curves.

Lemma 2.4 (The regular lift avoids the vertical divisor)**.** *Let $f : \mathbb{C} \to X_0$ be nonconstant and satisfy $f(\mathbb{C}) \not\subset D$. Then its logarithmic lift $f_{[2]}$ is well defined after holomorphic extension across $f^{-1}(D)$ and the isolated zeros of f', and*

$$f_{[2]}(\mathbb{C}) \not\subset \Gamma_2.$$

Proof. On the nonempty open set on which f avoids D and $f' \neq 0$, the first lift is $f_{[1]}(z) = (f(z), [f'(z)])$. Its derivative is tangent to V_1, and its projection by $d\pi_{1,0}$ is $f'(z) \neq 0$. Hence the line $[f'_{[1]}(z)] \subset V_{1,f_{[1]}(z)}$ is not vertical for $\pi_{1,0}$. By definition, Γ_2 consists precisely of the vertical lines at the second stage, so $f_{[2]}(z) \notin \Gamma_2$ on this open set. To extend the lifts, take a local coordinate τ at an omitted point. In a logarithmic frame, a normal component of the first derivative is $(w \circ f)'/(w \circ f)$, where $w = 0$ defines a local branch of D; the remaining components are holomorphic. These components form a nonzero meromorphic vector in $f^*\overline{T}$. Multiplying by a power of τ and removing their common vanishing factor gives a holomorphic vector nonzero at $\tau = 0$, hence a unique extension of its projective line. This extends $f_{[1]}$. In a local Semple chart its derivative has meromorphic horizontal logarithmic components and derivatives of holomorphic projective-coordinate ratios in the vertical components. It is therefore a nonzero meromorphic vector in $f^*_{[1]}V_1$: membership in V_1 holds on the punctured disk, and V_1 is a locally free subbundle. The same cancellation extends its projective line and thus $f_{[2]}$. The omitted set is discrete because f is nonconstant and is not contained in D. Since the complement of Γ_2 has already met the lifted curve, its whole image cannot be contained in Γ_2. □

2.4. **Jet Second Main Theorem and McQuillan's theorem.** We use the following standard jet-differential estimate; compare [2] and [6, Theorem 1.2]. The invariant logarithmic jet bundle $E_{k,m}T^*_{\mathbb{P}^2}(\log D)$ is naturally a subbundle of the Green–Griffiths bundle appearing in the statement, so the estimate applies to the invariant differentials constructed below.

Theorem 2.5 (Jet differential Second Main Theorem)**.** *Let $k, m, t \geqslant 1$ be integers. If*

$$0 \neq \omega \in H^0\big(\mathbb{P}^2, E^{\mathrm{GG}}_{k,m}T^*_{\mathbb{P}^2}(\log D) \otimes \mathcal{O}(-t)\big)$$

and $f : \mathbb{C} \to \mathbb{P}^2$ satisfies $f(\mathbb{C}) \not\subset D$ and $f^*\omega \not\equiv 0$, then

$$T_f(r) \leqslant \frac{m}{t} N^{[1]}_f(r, D) + O(\log T_f(r) + \log r) \ \|. \tag{2.8}$$

We also use McQuillan's algebraic-degeneracy theorem in the following quantitative form; see [13] and [6, Theorem 1.1].

Theorem 2.6 (McQuillan)**.** *Let $f : \mathbb{C} \to \mathbb{P}^2$ be algebraically nondegenerate. If its first logarithmic lift is algebraically degenerate, then for a normal-crossing divisor of total degree $d \geqslant 4$,*

$$(d-3)T_f(r) \leqslant N^{[1]}_f(r, D) + o(T_f(r)) \ \|.$$

For $d = s = d_1 + d_2$ this gives

$$T_f(r) \leqslant \frac{1}{s-3} N^{[1]}_f(r, D) + o(T_f(r)) \ \|. \tag{2.9}$$

2.5. **Cohomological and zero-locus inputs.** We next record the vanishing and weighted-degree estimates that produce the negatively twisted differentials used in the two main cases.

Lemma 2.7 (Logarithmic Bogomolov vanishing)**.** *Let $D = C_1 + C_2$ be a transverse pair of smooth plane curves, put $\kappa = d_1 + d_2 - 3 > 0$, $\overline{K} = K_{\mathbb{P}^2} + D$, and set $\overline{T} = T_{\mathbb{P}^2}(-\log D)$. For integers $p \geqslant 0$ and q,*

$$H^0\big(\mathbb{P}^2, S^p\overline{T} \otimes \overline{K}^q\big) = 0 \qquad \text{if} \qquad p > 2q.$$

Proof. This is the logarithmic Bogomolov-type vanishing theorem [6, Theorem 2.1]. □

Theorem 2.8 (The required logarithmic H^2 vanishing)**.** *Let $D = C_1 + C_2$ be a transverse pair of smooth plane curves with $\kappa = d_1 + d_2 - 3 > 0$, and put $\overline{K} = K_{\mathbb{P}^2} + D \simeq \mathcal{O}_{\mathbb{P}^2}(\kappa)$. For every rational number $0 \leqslant \delta < 1/3$ and every sufficiently divisible $m \gg 1$,*

$$H^2\Big(\mathbb{P}^2, E_{2,m}T^*_{\mathbb{P}^2}(\log D) \otimes \overline{K}^{-\delta m}\Big) = 0. \tag{2.10}$$

Proof. By Serre duality, the dual of the group in (2.10) is

$$H^0\Big(\mathbb{P}^2, K_{\mathbb{P}^2} \otimes (E_{2,m}\overline{T}^*)^\vee \otimes \overline{K}^{\delta m}\Big).$$

The dual of the invariant-two-jet filtration has graded pieces

$$S^{m-3j}\overline{T} \otimes \overline{K}^{-j} \otimes K_{\mathbb{P}^2} \otimes \overline{K}^{\delta m}, \qquad 0 \leqslant j \leqslant \lfloor m/3 \rfloor.$$

Because m is sufficiently divisible, δm is an integer. Multiplication by a nonzero section of $\mathcal{O}_{\mathbb{P}^2}(3)$ injects each graded piece into

$$S^{m-3j}\overline{T} \otimes \overline{K}^{\delta m - j}. \tag{2.11}$$

For (2.11), the inequality in Lemma 2.7 is equivalent to

$$(m-3j) - 2(\delta m - j) = (1-2\delta)m - j \geqslant (2/3 - 2\delta)m > 0.$$

Thus every graded piece has vanishing H^0. Induction along the filtration gives the vanishing of the whole Serre-dual group, proving (2.10). □

For the zero-locus argument, write

$$\bar{c}_i = c_i\big(T^*_{\mathbb{P}^2}(\log D)\big), \qquad h = c_1(\mathcal{O}_{\mathbb{P}^2}(1)),$$

and identify these classes with their pullbacks to the Semple tower. For $\rho = t/m$, set

$$Q_{\mathrm{DEG}}(\rho,\tau) = 3\tau^2 + \big(9\rho - 12(\bar{c}_1 \cdot h)\big)\tau + (13\bar{c}_1^2 - 9\bar{c}_2) - 12(\bar{c}_1 \cdot h)\rho. \tag{2.12}$$

When its value at $\tau = 0$ is positive, denote its smaller positive root by

$$\tau_1(\rho) = \frac{12\kappa - 9\rho - \sqrt{(12\kappa - 9\rho)^2 - 12\big((13\bar{c}_1^2 - 9\bar{c}_2) - 12\kappa\rho\big)}}{6}. \tag{2.13}$$

Section 3 derives these expressions from the two-component Chern classes and the tower intersection relations.

Lemma 2.9 (Tautological evaluation of a restricted section)**.** *Let $Z \subset X_2$ be an effective Cartier divisor and let*

$$\sigma \in H^0\big(Z, \mathcal{O}_{X_2}(2p,p) \otimes \pi_{2,0}^*\mathcal{O}_{\mathbb{P}^2}(-\widetilde{t})\big|_Z\big), \qquad p, \widetilde{t} \in \mathbb{Z}_{>0}.$$

Let $f\colon \mathbb{C} \to \mathbb{P}^2$ be nonconstant, assume $f(\mathbb{C}) \not\subset D$, and suppose that its regular logarithmic second lift is contained in Z. On $\mathbb{C} \setminus f^{-1}(D)$, regard $df_{[1]}\colon T_{\mathbb{C}} \to f_{[1]}^*V_1$ as a differential in the directed structure. Wherever it is nonzero, its image is the tautological line $f_{[2]}^*\mathcal{O}_{X_2}(-1)$. It therefore defines, by meromorphic continuation across the remaining discrete set, the intrinsic tautological derivative

$$\mathcal{D}_D f_{[1]} \in H^0_{\mathrm{mer}}\Big(\mathbb{C}, K_{\mathbb{C}} \otimes f_{[2]}^*\mathcal{O}_{X_2}(-1)\Big).$$

Then the canonical inclusion

$$\iota_p\colon \mathcal{O}_{X_2}(2p,p) = \mathcal{O}_{X_2}(3p)(-2p\Gamma_2) \hookrightarrow \mathcal{O}_{X_2}(3p)$$

defines the intrinsic meromorphic evaluation

$$s_{\sigma,f} := \Big\langle f_{[2]}^*(\iota_p\sigma), (\mathcal{D}_D f_{[1]})^{\otimes 3p}\Big\rangle \in H^0_{\mathrm{mer}}\Big(\mathbb{C}, K_{\mathbb{C}}^{3p} \otimes f^*\mathcal{O}_{\mathbb{P}^2}(-\widetilde{t})\Big). \tag{2.14}$$

This construction does not require an ambient lift of σ. Moreover,

$$\mathrm{div}_\infty(s_{\sigma,f}) \leqslant 3p\sum_{i=1}^{2}(f^*C_i)_{\mathrm{red}}, \tag{2.15}$$

and

$$\int_0^{2\pi} \log^+ \|s_{\sigma,f}(re^{\sqrt{-1}\theta})\| \frac{d\theta}{2\pi} = O\big(\log^+ T_f(r) + \log r\big) \ \|. \tag{2.16}$$

At an isolated zero of f' outside $f^{-1}(D)$, the evaluation has only a removable singularity or a zero. At a zero lying over D, no pole occurs beyond the logarithmic boundary poles already counted in (2.15).

Proof. Pairing its $3p$-th tensor power with $f_{[2]}^*(\iota_p\sigma)$ gives (2.14). Since this is a pairing of dual tautological lines, the result is independent of all Semple coordinates and all local representatives of σ.

Equip the line bundles with smooth Hermitian metrics. Since Z is projective, the norm of the holomorphic section $\iota_p\sigma$ is bounded on Z. Hence

$$\|s_{\sigma,f}\| \leqslant C\,\|\mathcal{D}_D f_{[1]}\|^{3p}$$

for a uniform constant C.

In coordinates in which $D = \{x_1 \cdots x_q = 0\}$, the horizontal boundary components of the derivative of the regular first lift are $(x_i \circ f)'/(x_i \circ f)$, while the vertical projective-coordinate components are holomorphic. Thus the tautological derivative has at most a simple pole along each reduced pullback $(f^*C_i)_{\mathrm{red}}$.

By Lemma 2.4, the Semple lift extends across the isolated zeros of f'. At a critical point outside $f^{-1}(D)$, the projective ratios extend and the evaluation has only a removable singularity or a zero. At a critical point over D, a logarithmic component may retain a simple pole; that pole is already counted by the reduced boundary pullback, and no additional Semple-type pole appears. This proves (2.15). Choose a finite logarithmic base cover and, above it, finitely many relatively compact Semple charts. A partition of unity is used only to estimate the norm of this single global section. On each chart, the components of $\mathcal{D}_D f_{[1]}$ are logarithmic derivatives of local coordinate functions. A projective Semple coordinate u is a rational function on X_1 and, in a logarithmic frame, a ratio of first logarithmic derivatives. Hence

$$T_{u\circ f_{[1]}}(r) = O\big(T_f(r) + \log r\big).$$

For a partition function χ supported in a relatively compact Semple chart, choose $a \notin \overline{u(\operatorname{supp}\chi)}$. On that support, $u - a$ is bounded above and away from zero, and

$$(u \circ f_{[1]})' = (u \circ f_{[1]} - a)\frac{(u \circ f_{[1]})'}{u \circ f_{[1]} - a}.$$

The logarithmic-derivative lemma applied to $u \circ f_{[1]} - a$ controls the vertical component by

$$O\big(\log^+ T_f(r) + \log r\big).$$

Summing over the finite cover, together with the analogous horizontal estimates, gives (2.16). This is the argument of [8, Thm. 3.1, proof of (3.3)–(3.6)], with the Semple-coordinate adaptation made explicit. □

Lemma 2.10 (Second Main Theorem for a section restricted to Z)**.** *Let $Z \subset X_2$ be an effective Cartier divisor and let*

$$\sigma \in H^0\big(Z, \mathcal{O}_{X_2}(2p,p) \otimes \pi_{2,0}^*\mathcal{O}_{\mathbb{P}^2}(-\widetilde{t})\big|_Z\big), \qquad p, \widetilde{t} \in \mathbb{Z}_{>0}.$$

Suppose that $f : \mathbb{C} \to \mathbb{P}^2$ is nonconstant, $f(\mathbb{C}) \not\subset D$, $f_{[2]}(\mathbb{C}) \subset Z$, and the canonical evaluation $s_{\sigma,f}$ of Lemma 2.9 is not identically zero. Then

$$\widetilde{t}\,T_f(r) \leqslant 3p\sum_{i=1}^{2} N_f^{[1]}(r, C_i) + O(\log T_f(r) + \log r)\ \|. \tag{2.17}$$

In particular, if f is algebraically nondegenerate, then

$$T_f(r) \leqslant \frac{3p}{\widetilde{t}} N_f^{[1]}(r, D) + o(T_f(r))\ \|.$$

Proof. By Lemma 2.9, the section $s_{\sigma,f}$ is globally and intrinsically defined, its pole divisor is bounded by (2.15), and its proximity term satisfies (2.16). Use the standard coordinate z on $\mathbb{C}$ to trivialize $K_{\mathbb{C}}^{3p}$ by the global frame $(dz)^{3p}$ and give it the flat metric. Equip $\mathcal{O}_{\mathbb{P}^2}(-1)$ with the metric dual to the Fubini–Study metric. Poincaré–Lelong and Jensen's formula give

$$\widetilde{t}\,T_f(r) \leqslant 3p\sum_{i=1}^{2} N_f^{[1]}(r, C_i) + O\big(\log^+ T_f(r) + \log r\big) \ \|.$$

This is (2.17). It is the intrinsic restricted-section version of the argument in [6, Proposition 6.3 and Theorem 6.4]; the analytic norm estimate is the one in [8, Theorem 3.1, proof of (3.3)–(3.6)]. □

Theorem 2.11 (Two-component zero-locus theorem and standard estimate)**.** *Let ω_1 be an irreducible nonzero section of*

$$E_{2,m}\overline{T}^{*} \otimes \mathcal{O}(-t), \qquad \rho = t/m,$$

whose zero divisor on X_2 is $Z + b\Gamma_2$, where Z is irreducible and reduced and

$$Z \sim b_1u_1 + b_2u_2 - th, \qquad b_1 \geqslant 2b_2 > 0, \qquad b_1 + b_2 = m. \tag{2.18}$$

Suppose

$$0 < \rho < \rho_{\mathrm{DEG}} := \frac{13\bar{c}_1^2 - 9\bar{c}_2}{12(\bar{c}_1 \cdot h)}.$$

Assume $\kappa = d_1 + d_2 - 3 \geqslant 3$. Let $\tau_1(\rho)$ be the smaller positive root of $Q_{\mathrm{DEG}}(\rho, \tau) = 0$, explicitly given in (2.13). For every rational

$$0 < \tau < \min\{3, \tau_1(\rho)\},$$

the $\mathbb{Q}$-line bundle

$$\big(\mathcal{O}_{X_2}(2,1) \otimes \pi_{2,0}^{*}\mathcal{O}_{\mathbb{P}^2}(-\tau)\big)\,|_Z$$

is big. Moreover, if $f : \mathbb{C} \to \mathbb{P}^2$ is algebraically nondegenerate and $f^{*}\omega_1 \equiv 0$, then either

$$T_f(r) \leqslant \frac{3}{\tau} N_f^{[1]}(r, D) + o(T_f(r)) \ \|, \tag{2.19}$$

or the first logarithmic lift $f_{[1]}$ is algebraically degenerate.

Proof. The restricted-bigness calculation follows the method of [6, Proposition 6.2], with the two-component intersection polynomial computed below. The local frame and analytic estimate are adapted from [6, Proposition 6.3 and Theorem 6.4] and are recorded in Lemma 2.10.

Put

$$L_\tau = \mathcal{O}_{X_2}(2,1) \otimes \pi_{2,0}^{*}\mathcal{O}_{\mathbb{P}^2}(-\tau).$$

The intersection computation on the logarithmic Demailly–Semple tower gives

$$c_1(L_\tau)^3 \cdot Z = mQ_{\mathrm{DEG}}(\rho, \tau). \tag{2.20}$$

By the definition of $\tau_1(\rho)$, the right-hand side is positive in the stated range.

We prove bigness without inferring it from (2.20) alone. Let $p \gg 1$ be sufficiently divisible, so that $p\tau/\kappa \in \mathbb{Z}$ (and hence $p\tau \in \mathbb{Z}$) and L_τ^p is an honest line bundle. Since X_2 is smooth and Z is an effective Cartier divisor, the exact sequence

$$0 \to L_\tau^p(-Z) \to L_\tau^p \to L_\tau^p|_Z \to 0$$

and Hirzebruch–Riemann–Roch on the smooth fourfold X_2 give

$$\chi(Z, L_\tau^p|_Z) = \chi(X_2, L_\tau^p) - \chi(X_2, L_\tau^p(-Z)) \tag{2.21}$$

$$= \int_{X_2} e^{pc_1(L_\tau)}\big(1 - e^{-[Z]}\big)\,\mathrm{td}(T_{X_2}) \tag{2.22}$$

$$= \frac{c_1(L_\tau)^3 \cdot Z}{3!}p^3 + O(p^2). \tag{2.23}$$

This ambient computation does not require Z to be smooth or normal.

We next show $H^2(Z, L_\tau^p|_Z) = 0$. Lemma 2.3, the projection formula, and Leray yield

$$H^2(X_2, L_\tau^p) \simeq H^2\Big(\mathbb{P}^2, E_{2,3p}\overline{T}^* \otimes \mathcal{O}(-p\tau)\Big) = 0,$$

where Theorem 2.8 is applied with $\delta = \tau/(3\kappa) < 1/3$.

Using (2.18),

$$L_\tau^p(-Z) \simeq \mathcal{O}_{X_2}(2p - b_1, p - b_2) \otimes \pi_{2,0}^*\mathcal{O}_{\mathbb{P}^2}(t - p\tau).$$

The tempting attempt to apply the preceding relative vanishing directly to this bundle is incorrect in general: on the fiber $\mathbb{F}_3$ one must control $a_1 - 2a_2$, and here that number equals $2b_2 - b_1$, which may be arbitrarily negative. We instead prove the required H^3 vanishing by a filtration.

Write $q = \pi_{2,1} : X_2 \to X_1$ and $\pi = \pi_{1,0} : X_1 \to \mathbb{P}^2$. For $p \gg 1$ one has $p - b_2 \geqslant 0$, whence

$$R^r q_* L_\tau^p(-Z) = 0 \quad (r \geqslant 1)$$

and

$$q_* L_\tau^p(-Z) = \mathcal{O}_{X_1}(2p - b_1) \otimes \mathrm{Sym}^{p-b_2} V_1^* \otimes \pi^*\mathcal{O}(t - p\tau).$$

The dual relative tangent sequence

$$0 \longrightarrow \mathcal{O}_{X_1}(1) \longrightarrow V_1^* \longrightarrow \pi^*\overline{K} \otimes \mathcal{O}_{X_1}(-2) \longrightarrow 0 \tag{2.24}$$

induces a filtration whose quotients, indexed by $0 \leqslant j \leqslant p - b_2$, are

$$\mathcal{O}_{X_1}(3p - m - 3j) \otimes \pi^*\Big(\overline{K}^j \otimes \mathcal{O}(t - p\tau)\Big).$$

Put $\ell = p - j$. The same quotients become

$$\mathcal{G}_{\ell,p} = \mathcal{O}_{X_1}(3\ell - m) \otimes \pi^*\mathcal{O}\big((\kappa - \tau)p + t - \kappa\ell\big), \qquad b_2 \leqslant \ell \leqslant p. \tag{2.25}$$

If $3\ell - m \geqslant -1$, then $R^1\pi_*\mathcal{O}_{X_1}(3\ell - m) = 0$, so $H^3(X_1, \mathcal{G}_{\ell,p}) = 0$ by Leray. The remaining indices satisfy

$$b_2 \leqslant \ell \leqslant \left\lfloor \frac{m-2}{3} \right\rfloor;$$

this is a fixed finite set, independent of p. For each such ℓ, the projection formula gives

$$R^1\pi_*\mathcal{G}_{\ell,p} = R^1\pi_*\mathcal{O}_{X_1}(3\ell - m) \otimes \mathcal{O}\big((\kappa - \tau)p + t - \kappa\ell\big).$$

The first factor is fixed, while $\kappa - \tau > 0$ because $\tau < 3 \leqslant \kappa$. Serre vanishing therefore kills its H^2 for $p \gg 1$. Leray, followed by induction through the filtration (2.24), yields

$$H^3(X_2, L_\tau^p(-Z)) = 0. \tag{2.26}$$

The divisor exact sequence now gives $H^2(Z, L_\tau^p|_Z) = 0$. Combining this with (2.21), we obtain

$$h^0(Z, L_\tau^p|_Z) = \chi(Z, L_\tau^p|_Z) + h^1(Z, L_\tau^p|_Z) + h^3(Z, L_\tau^p|_Z) \geqslant \chi(Z, L_\tau^p|_Z) \gg p^3,$$

which proves that the $\mathbb{Q}$-line bundle $L_\tau|_Z$ is big.

Choose p in the same sufficiently divisible sequence and a nonzero section

$$\sigma \in H^0(Z, L_\tau^p|_Z), \qquad \widetilde{t} = p\tau \in \mathbb{Z}.$$

Since $f^*\omega_1 \equiv 0$ and Lemma 2.4 excludes containment in Γ_2, one has $f_{[2]}(\mathbb{C}) \subset Z$. Suppose first that the canonical evaluation $s_{\sigma,f}$ is not identically zero. Since

$$L_\tau^p = \mathcal{O}_{X_2}(2p, p) \otimes \pi_{2,0}^*\mathcal{O}_{\mathbb{P}^2}(-\widetilde{t}),$$

Lemma 2.10 yields

$$\widetilde{t}\, T_f(r) \leqslant 3pN_f^{[1]}(r, D) + O(\log T_f(r) + \log r) \ \|.$$

The logarithmic derivative lemma makes the final term $o(T_f(r))$ for an algebraically nondegenerate curve. Since $\widetilde{t} = p\tau$, this is (2.19).

If instead $s_{\sigma,f} \equiv 0$, the tautological derivative is nonzero away from a discrete set because f is nonconstant. The intrinsic pairing (2.14) therefore implies $f_{[2]}^*\sigma \equiv 0$. Hence $f_{[2]}(\mathbb{C})$ is

contained in the proper divisor $\operatorname{div}_Z(\sigma)$ of the irreducible threefold Z. This set has dimension at most two, so its image under $\pi_{2,1} : X_2 \to X_1$ is a proper algebraic subset of the threefold X_1. Hence $f_{[1]}$ is algebraically degenerate. □

Remark 2.12. The preceding proof keeps separate the three ingredients required by the argument: the positive cubic intersection, the H^2 and H^3 vanishings needed to obtain bigness, and the intrinsic tautological evaluation and logarithmic-derivative estimate used in (2.19).

2.6. **Algebraically degenerate curves.** The preceding analytic estimates concern algebraically nondegenerate curves. For the hyperbolicity corollary, the algebraically degenerate case is supplied by the following specialization of Xi Chen's theorem.

For an irreducible curve $\Gamma \not\subset D$, let $\nu : \widetilde{\Gamma} \to \Gamma$ be the normalization and define

$$i(\Gamma, D) := \#\operatorname{Supp} \nu^* D.$$

Theorem 2.13 (Xi Chen)**.** *For a very general ordered pair of smooth transverse plane curves of total degree $s = d_1 + d_2$ and every reduced irreducible curve $\Gamma \not\subset D$,*

$$2g(\widetilde{\Gamma}) - 2 + i(\Gamma, D) \geqslant (s-4)\deg \Gamma. \tag{2.27}$$

Source of the specialization. Xi Chen proves for a very general reducible plane divisor $D = \bigcup D_j$ of total degree d that

$$2g(\widetilde{\Gamma}) - 2 + i(\Gamma, D) \geqslant (d-4)\deg \Gamma.$$

Taking a very general ordered pair of degrees (d_1, d_2) gives $d = s$ and hence (2.27); see [1, Corollary 1.11]. □

Corollary 2.14. *If $s \geqslant 5$, then for a very general pair there is no nonconstant algebraically degenerate entire curve in $\mathbb{P}^2 \setminus D$.*

Proof. If $f(\mathbb{C})$ is contained in $\Gamma \not\subset D$, it lifts to

$$\widetilde{f} : \mathbb{C} \longrightarrow \widetilde{\Gamma} \setminus \operatorname{Supp} \nu^* D.$$

By (2.27),

$$2g(\widetilde{\Gamma}) - 2 + i(\Gamma, D) > 0,$$

so the punctured Riemann surface on the right is hyperbolic. Hence $\widetilde{f}$ and f are constant. □

3. Logarithmic Chern numbers and the degree phase diagram

Two ratios organize the proof. The Riemann–Roch threshold ρ_{RR} permits the initial choice $t/m < \rho_{\mathrm{RR}}$; the Demailly–El Goul threshold ρ_{DEG} determines when an irreducible factor lies in the zero-locus range. Keeping both degrees symbolic isolates the six pairs for which the remaining small twists require finite verification.

3.1. **The two-component Chern data.** For $h = c_1(\mathcal{O}_{\mathbb{P}^2}(1))$ one has

$$\overline{K} = K_{\mathbb{P}^2} + D \simeq \mathcal{O}_{\mathbb{P}^2}(\kappa), \qquad \bar{c}_1 = \kappa h, \qquad \kappa = d_1 + d_2 - 3.$$

Using

$$c\big(\Omega^1_{\mathbb{P}^2}(\log D)\big) = c(\Omega^1_{\mathbb{P}^2}) \prod_{i=1}^{2} (1 + [C_i] + [C_i]^2)$$

and $[C_i] = d_i h$, we obtain

$$\bar{c}_1^2 = \kappa^2, \qquad \bar{c}_2 = d_1^2 + d_2^2 + d_1 d_2 - 3(d_1 + d_2) + 3, \qquad \bar{c}_1 \cdot h = \kappa. \tag{3.1}$$

Two combinations will recur:

$$\bar{c}_1^2 - \bar{c}_2 = d_1 d_2 - 3(d_1 + d_2) + 6, \tag{3.2}$$

$$\begin{aligned} 13\bar{c}_1^2 - 9\bar{c}_2 &= A(d_1, d_2) \\ &= 4(d_1^2 + d_2^2) + 17 d_1 d_2 - 51(d_1 + d_2) + 90. \end{aligned} \tag{3.3}$$

The first is positive in the classical one-jet range $d_1 \geqslant 5$, or $d_1 = 4, d_2 \geqslant 7$. The second is positive for every $d_1, d_2 \geqslant 3$; its minimum there is $A(3,3) = 9$. On the degree-two boundary,

$$A(2,n) = 4n^2 - 17n + 4, \tag{3.4}$$

which is positive exactly for integers $n \geqslant 5$. On the degree-one boundary,

$$A(1,n) = 4n^2 - 34n + 43, \tag{3.5}$$

which, for integers $n \geqslant 2$, is positive exactly when $n \geqslant 7$.

3.2. **Riemann–Roch threshold.** This is the standard invariant-jet Riemann–Roch calculation introduced in Demailly's foundational note [2]. We spell out only the two-component specialization of the leading coefficient because it determines the effective range used later. By (2.2), additivity of the Chern character gives

$$\chi\Big(E_{2,m}\overline{T}^* \otimes \overline{K}^{-\delta m}\Big) = \sum_{j=0}^{\lfloor m/3 \rfloor} \chi\Big(S^{m-3j}\overline{T}^* \otimes \overline{K}^{j-\delta m}\Big). \tag{3.6}$$

Let x_1, x_2 be the formal Chern roots of $\overline{T}^*$. For $q = m - 3j$,

$$\mathrm{ch}(S^q\overline{T}^*) = \sum_{a=0}^{q} e^{ax_1 + (q-a)x_2}.$$

Keeping terms of cohomological degree two, multiplying by $e^{(j-\delta m)(x_1+x_2)}\,\mathrm{td}(T_{\mathbb{P}^2})$, and integrating over $\mathbb{P}^2$ computes each summand in (3.6). The Todd terms of positive degree contribute only $O(m^3)$ after summation. For the degree-four term, the elementary identities

$$\sum_{a=0}^{q} 1 = q+1, \qquad \sum_{a=0}^{q} a = \frac{q(q+1)}{2}, \qquad \sum_{a=0}^{q} a^2 = \frac{q(q+1)(2q+1)}{6}$$

followed by the sums in $0 \leqslant j \leqslant \lfloor m/3 \rfloor$ give

$$\begin{aligned} &\sum_{j=0}^{\lfloor m/3 \rfloor} \int_{\mathbb{P}^2} \Big[\mathrm{ch}(S^{m-3j}\overline{T}^*) e^{(j-\delta m)\bar{c}_1}\Big]_2 \\ &\qquad = \frac{m^4}{648}\left((54\delta^2 - 48\delta + 13)\bar{c}_1^2 - 9\bar{c}_2\right) + O(m^3). \end{aligned} \tag{3.7}$$

Define

$$Q_{\mathrm{RR}}(\delta) = 54\bar{c}_1^2\delta^2 - 48\bar{c}_1^2\delta + (13\bar{c}_1^2 - 9\bar{c}_2).$$

Then, for $t = (\bar{c}_1 \cdot h)\delta m$,

$$\chi\Big(\mathbb{P}^2, E_{2,m}\overline{T}^* \otimes \overline{K}^{-\delta m}\Big) = \frac{Q_{\mathrm{RR}}(\delta)}{648} m^4 + O(m^3). \tag{3.8}$$

Thus the leading coefficient is positive for sufficiently small positive δ. In the ratio $\rho = t/m = \kappa\delta$, its smaller positive root is

$$\rho_{\mathrm{RR}}(d_1, d_2) = \frac{\kappa}{18}\left(8 - \sqrt{54\frac{\bar{c}_2}{\bar{c}_1^2} - 14}\right) > 0. \tag{3.9}$$

The strict positivity is equivalent to $A(d_1, d_2) > 0$. For two cubics this specializes to $(8 - \sqrt{58})/6 > 1/16$.

3.3. **Zero-locus threshold.** Set $\rho = t/m$. Using the projective-bundle relations on the logarithmic Demailly–Semple tower, the cubic intersection occurring in Theorem 2.11 is

$$(2u_1 + u_2 - \tau h)^3 \cdot Z = mQ_{\mathrm{DEG}}(\rho, \tau), \tag{3.10}$$

where, as in (2.12),

$$Q_{\mathrm{DEG}}(\rho, \tau) = 3\tau^2 + (9\rho - 12(\bar{c}_1 \cdot h))\tau + (13\bar{c}_1^2 - 9\bar{c}_2) - 12(\bar{c}_1 \cdot h)\rho.$$

Here is the intersection calculation. Put $\Theta = 2u_1 + u_2$ and write $[Z] = b_1u_1 + b_2u_2 - th$, where $b_1 + b_2 = m$. Apply successively the rank-two projective-bundle formula

$$\pi_*(u) = 1, \qquad \pi_*(u^2) = -c_1(V), \qquad \pi_*(u^3) = c_1(V)^2 - c_2(V)$$

first to $\pi_{2,1}$ and then to $\pi_{1,0}$. With our convention that $\mathbb{P}(V)$ parametrizes lines, the intermediate identities are

$$u_1^2 - \bar{c}_1u_1 + \bar{c}_2 = 0, \qquad c_1(V_1) = u_1 - \bar{c}_1, \qquad c_2(V_1) = 2\bar{c}_2 - \bar{c}_1u_1.$$

Expanding with these relations gives the following three identities on the base:

$$\Theta h^2 \cdot Z = m, \tag{3.11}$$

$$\Theta^2 h \cdot Z = 4m(\bar{c}_1 \cdot h) - 3t, \tag{3.12}$$

$$\Theta^3 \cdot Z = m(13\bar{c}_1^2 - 9\bar{c}_2) - 12t(\bar{c}_1 \cdot h). \tag{3.13}$$

The terms involving b_1 and b_2 combine through $b_1 + b_2 = m$; this is why the answer is independent of the chosen irreducible factor. Since $h^3 = 0$ on the surface base, expansion of $(\Theta - \tau h)^3 \cdot Z$ and substitution of (3.11)–(3.13) yield

$$\begin{aligned}(\Theta - \tau h)^3 \cdot Z &= \Theta^3 \cdot Z - 3\tau\Theta^2 h \cdot Z + 3\tau^2\Theta h^2 \cdot Z \\ &= mQ_{\mathrm{DEG}}(\rho, \tau),\end{aligned}$$

which proves (3.10) directly. At $\tau = 0$ this is positive exactly when

$$\rho < \rho_{\mathrm{DEG}} = \frac{13\bar{c}_1^2 - 9\bar{c}_2}{12(\bar{c}_1 \cdot h)}. \tag{3.14}$$

Using (3.3),

$$\rho_{\mathrm{DEG}}(d_1, d_2) = \frac{A(d_1, d_2)}{12\kappa}. \tag{3.15}$$

Write $A = A(d_1, d_2)$. In the stated degree ranges, $\kappa \geqslant 3$ and $A > 0$. Formula (3.3) gives

$$\frac{32}{3}\kappa^2 - A = \frac{53\kappa^2 + 162\kappa + 81 + 27(d_1 - d_2)^2}{12} > 0.$$

For $0 \leqslant \rho < \rho_{\mathrm{DEG}} = A/(12\kappa)$, one therefore has

$$Q_{\mathrm{DEG}}(\rho, 0) = A - 12\kappa\rho > 0, \qquad Q_{\mathrm{DEG}}\left(\rho, \frac{4\kappa}{3}\right) = A - \frac{32\kappa^2}{3} < 0.$$

Thus the quadratic has two distinct positive roots, and its smaller root (2.13) satisfies $0 < \tau_1(\rho) < 4\kappa/3$. At this root $\partial_\tau Q_{\mathrm{DEG}} < 0$, so implicit differentiation gives

$$\tau_1'(\rho) = \frac{12\kappa - 9\tau_1(\rho)}{6\tau_1(\rho) + 9\rho - 12\kappa} < 0. \tag{3.16}$$

In particular, $3/\tau_1(\rho)$ is strictly increasing on $[0, \rho_{\mathrm{DEG}})$. For $(3, 3)$ this is

$$\tau_1(\rho) = 6 - \frac{3}{2}\rho - \frac{1}{2}\sqrt{132 - 24\rho + 9\rho^2}. \tag{3.17}$$

Equation (3.10) supplies the positive leading term; bigness and the second differential follow from the cohomological theorem, not from this number alone.

3.4. **Why the proved range has six finite boundary pairs.** For a genuine second-level component one has $b_2 > 0$ and $b_1 \geqslant 2b_2$, hence $m = b_1 + b_2 \geqslant 3$. If $t \leqslant 3$, then $t/m \leqslant 1$. Consequently no finite vanishing is needed whenever $\rho_{\mathrm{DEG}} > 1$. We apply this observation only after the Riemann–Roch positivity condition $A(d_1, d_2) > 0$ has supplied the initial differential. Direct calculation gives

$$\rho_{\mathrm{DEG}}(3,3) = \frac{1}{4}, \qquad \rho_{\mathrm{DEG}}(3,4) = \frac{37}{48}, \qquad \rho_{\mathrm{DEG}}(3,5) = \frac{73}{60} > 1,$$

and $\rho_{\mathrm{DEG}} > 1$ for every other pair with $3 \leqslant d_1 \leqslant d_2$, except $(3,3)$ and $(3,4)$. On the degree-two boundary,

$$\rho_{\mathrm{DEG}}(2,5) = \frac{19}{48}, \qquad \rho_{\mathrm{DEG}}(2,6) = \frac{23}{30},$$

whereas $\rho_{\mathrm{DEG}}(2,n) > 1$ for $n \geqslant 7$. This proves the phase diagram on the degree-two boundary. Finally,

$$\rho_{\mathrm{DEG}}(1,8) = \frac{3}{8}, \qquad \rho_{\mathrm{DEG}}(1,9) = \frac{61}{84}, \qquad \rho_{\mathrm{DEG}}(1,10) = \frac{103}{96} > 1,$$

and $\rho_{\mathrm{DEG}}(1,n) > 1$ for every $n \geqslant 10$. Together these calculations reduce the asserted line-component range $n \geqslant 8$ to $(1,8)$ and $(1,9)$. The omitted formal endpoint $(1,7)$ has $A(1,7) = 1$ and $\rho_{\mathrm{DEG}}(1,7) = 1/60$; it is not a seventh certified boundary pair, because its required vanishing profile is neither proved nor claimed here. Remark 1.1 records that separate problem. Thus, within the theorem's stated range, the pairs with $A(d_1,d_2) > 0$ but $\rho_{\mathrm{DEG}} \leqslant 1$ are exactly $(3,3), (3,4), (2,5), (2,6), (1,8), (1,9)$. These are precisely the six finite lists in Lemma 1.2.

4. The first negatively twisted differential

The construction has three steps. Riemann–Roch and H^2 vanishing first give one section with a degree pair independent of the parameter. The Demailly–El Goul reduction then selects a prime horizontal component with no loss in the twist-to-weight ratio. Finally, cohomology and base change spread its canonical reduced equation over one Zariski-open subset of the ordered parameter space. The output is therefore a fixed numerical type (m, t, b_1, b_2) and one algebraic family of prime horizontal divisors; these are exactly the uniform data used by the mixed-field construction.

Proposition 4.1 (Uniform existence)**.** *Fix a rational number*

$$0 < \rho_0 < \rho_{\mathrm{RR}}(d_1, d_2). \tag{4.1}$$

There exist fixed positive integers M, T with $T/M = \rho_0$ such that for every $a \in \mathcal{S}^{\mathrm{snc}}_{d_1,d_2}$ one has

$$H^0\big(\mathbb{P}^2, E_{2,M}T^*_{\mathbb{P}^2}(\log D_a) \otimes \mathcal{O}_{\mathbb{P}^2}(-T)\big) \neq 0. \tag{4.2}$$

Proof. Put $\delta_0 = \rho_0/(\bar{c}_1 \cdot h) = \rho_0/\kappa$. Write $\rho_0 = T_0/M_0$ with positive integers M_0, T_0. Choose a sufficiently large N such that NT_0/κ is integral, and set

$$M = NM_0, \qquad T = NT_0.$$

Then $\delta_0 M = T/\kappa \in \mathbb{Z}$, so the twist $\overline{K}^{-\delta_0 M}$ is an honest line bundle. All smooth transverse pairs of degrees (d_1, d_2) have the same logarithmic Chern numbers (3.1). Hence the Riemann–Roch polynomial in (3.8), including its lower-order coefficients, is independent of the parameter $a \in \mathcal{S}^{\mathrm{snc}}_{d_1,d_2}$. Since $Q_{\mathrm{RR}}(\delta_0) > 0$, one may choose N once and for all so that this Euler characteristic is positive for every such a. Theorem 2.8, applied with the same δ_0, also holds for every D_a. Therefore

$$h^0 = h^1 + \chi - h^2 \geqslant \chi > 0$$

for every $a \in \mathcal{S}^{\mathrm{snc}}_{d_1,d_2}$, proving (4.2). □

The following irreducible-component reduction is due to Demailly–El Goul [3, Lemma 3.3]; we use the implementation in [6, Section 4]. The reduction itself is unchanged in the two-component setting. We reproduce the short argument only to fix the notation and the ratio inequality needed below.

Lemma 4.2 (Irreducible-component reduction)**.** *From a nonzero section of weighted degree M and twist T one may choose an irreducible horizontal factor*

$$0 \neq \omega_1 \in H^0\Big(\mathbb{P}^2, E_{2,m}\overline{T}^* \otimes \mathcal{O}(-t)\Big)$$

whose zero divisor on X_2 has the form $Z + b\Gamma_2$, with Z irreducible, reduced, and horizontal, and such that

$$\frac{t}{m} \geqslant \frac{T}{M}.$$

Writing the horizontal component as

$$Z \sim b_1u_1 + b_2u_2 - th, \qquad b_1 \geqslant 2b_2 \geqslant 0, \qquad b_1 + b_2 = m,$$

one has the following alternative: if $b_2 = 0$, then Z is pulled back from a divisor on X_1; if $b_2 > 0$, it is a genuine second-level horizontal divisor. In particular, applying the lemma to the fixed pair (M, T) supplied by Proposition 4.1 yields an irreducible ω_1 satisfying

$$\frac{t}{m} \geqslant \rho_0, \qquad \text{and hence} \qquad \frac{m}{t} \leqslant \rho_0^{-1}.$$

Proof. Via (2.3), the original section is represented on X_2 by an effective divisor

$$Z_\omega \sim Mu_2 - Th.$$

Write its irreducible decomposition as

$$Z_\omega = \sum_j p_j Z_j, \qquad Z_j \sim b_{1,j}u_1 + b_{2,j}u_2 - t_j h.$$

The standard classification of effective divisors on the second Demailly–Semple tower gives three possibilities:

(i) $(b_{1,j}, b_{2,j}) = (0, 0)$ and Z_j is pulled back from the base; then $t_j \leqslant 0$;
(ii) $(b_{1,j}, b_{2,j}) = (-1, 1)$, $t_j = 0$, and $Z_j = \Gamma_2$;
(iii) Z_j is horizontal and $b_{1,j} \geqslant 2b_{2,j} \geqslant 0$, with $b_{1,j} + b_{2,j} > 0$.

For a horizontal component put

$$m_j = b_{1,j} + b_{2,j} > 0.$$

Since the class of Z_ω has $u_1 + u_2$-weight M, while the first two types have weight zero, comparison of divisor classes gives

$$\sum_{j \text{ horizontal}} p_j m_j = M.$$

Comparison of the base twists gives

$$\sum_j p_j t_j = T.$$

The base-pullback components have $t_j \leqslant 0$ and Γ_2 has $t_j = 0$, so

$$\sum_{j \text{ horizontal}} p_j t_j \geqslant T.$$

Consequently at least one horizontal component satisfies

$$\frac{t_j}{m_j} \geqslant \frac{T}{M}.$$

Fix such a component and rename it Z. Because $\mathcal{O}_{X_2}(\Gamma_2) = \mathcal{O}_{X_2}(-1, 1)$, one has

$$Z + b_{1,j}\Gamma_2 \sim (b_{1,j} + b_{2,j})u_2 - t_j h = m_j u_2 - t_j h.$$

Thus, if s_j is the canonical reduced equation of the prime Cartier divisor Z and γ is the canonical section of Γ_2, then $\gamma^{b_{1,j}} s_j$ is a section of $\mathcal{O}_{X_2}(m_j) \otimes \pi_{2,0}^* \mathcal{O}(-t_j)$, which corresponds by (2.3) to an invariant jet differential

$$\omega_j \in H^0\Big(\mathbb{P}^2, E_{2,m_j}\overline{T}^* \otimes \mathcal{O}(-t_j)\Big).$$

After saturation by the fixed vertical factor $\gamma^{b_{1,j}}$, its horizontal zero divisor is exactly the reduced prime Z; no extraction of a root of a global section is involved. When $b_{2,j} = 0$, the classification identifies Z_j with the inverse image under $\pi_{2,1}$ of a divisor on X_1. The ratio inequality proves the final assertion. □

The following family-extension step follows the algebraic spread-out construction in [6, Section 4], applied to the ordered two-component parameter space. The only change is the base: the generic freeness, cohomology-and-base-change, and geometric-integrality arguments are the same.

Proposition 4.3 (Algebraic extension of the chosen factor)**.** *Let (M, T) be the fixed integers supplied by Proposition 4.1. Then there exist integers (m, t) with $t/m \geqslant T/M$, integers*

$$b_1 \geqslant 2b_2 \geqslant 0, \qquad b_1 + b_2 = m,$$

and a nonempty Zariski-open subset $\mathcal{U} \subset \mathcal{S}^{\mathrm{snc}}_{d_1,d_2}$ with the following property. Let

$$p_{\mathcal{U}} : \mathcal{X}_2 \longrightarrow \mathcal{U}, \qquad \Pi : \mathcal{X}_2 \longrightarrow \mathbb{P}^2 \times \mathcal{U}$$

be the relative second logarithmic Semple tower and its projection to level zero. There is a regular algebraic section

$$\int \in H^0(\mathcal{X}_2, \mathcal{L}), \qquad \mathcal{L} = \mathcal{O}_{\mathcal{X}_2}(b_1, b_2) \otimes \Pi^* \operatorname{pr}_1^* \mathcal{O}_{\mathbb{P}^2}(-t),$$

whose zero scheme $\mathcal{Z} = (\int = 0)$ is flat over $\mathcal{U}$ and has geometrically integral fibers. If γ denotes the canonical section of the relative vertical divisor Γ_2, then

$$\omega_{1,a} = \gamma_a^{b_1} s_a \in H^0\big(\mathbb{P}^2, E_{2,m} T^*_{\mathbb{P}^2}(\log D_a) \otimes \mathcal{O}_{\mathbb{P}^2}(-t)\big), \qquad a \in \mathcal{U},$$

is an algebraic family of invariant jet differentials, and its horizontal zero divisor is the irreducible reduced divisor $Z_a = \mathcal{Z}|_{X_{2,a}}$.

Proof. Work first over $S = \mathcal{S}^{\mathrm{snc}}_{d_1,d_2}$, and denote its relative second logarithmic Semple tower by $p_S : \mathcal{X}_{2,S} \to S$. For every triple (b_1, b_2, t) consider the algebraic line bundle

$$\mathcal{L}_{b_1,b_2,t} = \mathcal{O}_{\mathcal{X}_{2,S}}(b_1, b_2) \otimes \Pi_S^* \operatorname{pr}_1^* \mathcal{O}_{\mathbb{P}^2}(-t).$$

The sheaf $(p_S)_* \mathcal{L}_{b_1,b_2,t}$ is coherent. Generic freeness and generic cohomology and base change give a dense Zariski-open set $S_{b_1,b_2,t} \subset S$ on which it is locally free and the natural map

$$((p_S)_* \mathcal{L}_{b_1,b_2,t}) \otimes k(a) \longrightarrow H^0(X_{2,a}, \mathcal{L}_{b_1,b_2,t}|_{X_{2,a}}) \tag{4.3}$$

is an isomorphism. There are only countably many triples. Since the base field is the uncountable field $\mathbb{C}$, an irreducible variety of finite type over $\mathbb{C}$ cannot be the union of countably many proper closed subsets. Equivalently, every countable intersection of dense Zariski-open subsets has a complex point. We may therefore choose

$$a_0 \in \bigcap_{b_1,b_2,t} S_{b_1,b_2,t}.$$

This countable intersection is used only to select one numerical type and one prime factor; after that choice, the construction is spread over a single Zariski-open neighborhood. By Proposition 4.1, this fiber carries a section of the fixed weighted degree M and fixed twist T. Apply Lemma 4.2 on the fiber over a_0 and choose its reduced horizontal equation

$$0 \neq s_{a_0} \in H^0(X_{2,a_0}, \mathcal{L}_{b_1,b_2,t}|_{X_{2,a_0}})$$

with $b_1 \geqslant 2b_2 \geqslant 0$, $b_1 + b_2 = m$, and $t/m \geqslant T/M$. For this particular triple, the base-change isomorphism above identifies s_{a_0} with a vector in the fiber of the locally free sheaf $(p_S)_* \mathcal{L}_{b_1,b_2,t}$.

After shrinking to an affine trivializing neighbourhood, extend that vector to a regular local section of the pushforward. The adjunction evaluation map

$$p_S^*(p_S)_*\mathcal{L}_{b_1,b_2,t} \longrightarrow \mathcal{L}_{b_1,b_2,t}$$

then produces a regular section $\int$ on the relative tower after replacing S by an affine Zariski-open neighborhood of a_0.

The section s_{a_0} is nonzero and its zero scheme is the selected irreducible reduced horizontal divisor. Thus it is an integral Cartier divisor; over the algebraically closed field $\mathbb{C}$ it is also geometrically integral. Nonvanishing on every irreducible component of a fiber is an open condition, so after shrinking, $\mathcal{Z} = (\int = 0)$ is a relative effective Cartier divisor. It is then flat over S because both $\mathcal{X}_{2,S}$ and the invertible ideal $\mathcal{L}^{-1}$ are flat and

$$0 \longrightarrow \mathcal{L}^{-1} \xrightarrow{\int} \mathcal{O}_{\mathcal{X}_{2,S}} \longrightarrow \mathcal{O}_{\mathcal{Z}} \longrightarrow 0$$

remains exact on the fibers. Finally, geometric integrality of fibers is open for a proper flat morphism of finite presentation. A second shrinking therefore makes every Z_a geometrically integral. The relative divisor Γ_2 and its canonical section γ are algebraic. Consequently $\gamma^{b_1}\int$ is a regular relative section of

$$\mathcal{O}_{\mathcal{X}_2}(m) \otimes \Pi^* \operatorname{pr}_1^* \mathcal{O}_{\mathbb{P}^2}(-t),$$

and the relative direct-image formula gives the asserted algebraic family $\omega_{1,a}$. This proves all statements after naming the resulting open set $\mathcal{U}$. □

From now on we fix the irreducible differential supplied by Lemma 4.2 and denote its weighted degree and twist by (m, t).

5. Mixed $\mathcal{O}_{\mathbb{P}^2}(3)$ slanted vector fields

We follow the logarithmic low-pole construction: first construct matrix and quadratic-acceleration fields, then absorb their parameter components at a fixed point by third differences. For two components, the same acceleration must satisfy two Hessian equations. We solve these equations simultaneously and use a commutator of the resulting fields to obtain a second equation independent of the first one.

The coefficient differences and matrix lifts are based on [14, Section 3], [16, Lemma 1.1], and [17, Lemma 5]. The quadratic-acceleration method and its organization by symmetry are related to [9]. We give the calculations for two components, including the degree-one case and the common base twist $\mathcal{O}_{\mathbb{P}^2}(3)$.

5.1. **The auxiliary logarithmic family and the tangency test.** On $Z_0 \neq 0$, write the two universal equations as

$$F_i(z, a_i) = \sum_{|\alpha| \leqslant d_i} a_{i,\alpha} z^\alpha, \qquad z = (z_1, z_2), \qquad i = 1, 2.$$

We initially allow the two nonzero coefficient vectors a_1, a_2 to vary on their affine cones. The auxiliary family $\mathcal{Y}$ is given on these charts by

$$z_3^{d_1} + F_1(z, a_1) = 0, \qquad z_4^{d_2} + F_2(z, a_2) = 0. \tag{5.1}$$

Let π forget z_3, z_4 and project each coefficient vector to its projective class. The reduced inverse image of the universal divisor is $\{z_3 z_4 = 0\}$. On fixed coefficient charts, π is a product of cyclic covers; the corresponding changes of base and coefficient charts are checked below. At a crossing, F_1, F_2 are local coordinates and

$$\mathrm{d}\log F_i = \frac{\mathrm{d}F_i}{F_i} = \frac{-d_i z_{i+2}^{d_i-1}\mathrm{d}z_{i+2}}{-z_{i+2}^{d_i}} = d_i\,\mathrm{d}\log z_{i+2}.$$

Thus, over the SNC parameter locus the auxiliary family is smooth and its relative logarithmic jets describe the logarithmic jets downstairs. For $d_i = 1$ the corresponding equation is simply a graph.

As in the one-component case, for $r = 1, 2$ set

$$\xi_j^{(r)} = z_j^{(r)} \quad (j = 1, 2), \qquad \xi_{i+2}^{(r)} = (\log z_{i+2})^{(r)} \quad (i = 1, 2), \qquad \xi^{(r)} = (\xi_1^{(r)}, \xi_2^{(r)}).$$

The equations defining $\overline{J_2^v}(\mathcal{Y})$ are

$$E_{i,0}^{\log} = z_{i+2}^{d_i} + F_i = 0, \tag{5.2}$$

$$E_{i,1}^{\log} = d_i z_{i+2}^{d_i} \xi_{i+2}^{(1)} + \mathrm{d}F_i(\xi^{(1)}) = 0, \tag{5.3}$$

$$E_{i,2}^{\log} = d_i z_{i+2}^{d_i} \xi_{i+2}^{(2)} + d_i^2 z_{i+2}^{d_i} (\xi_{i+2}^{(1)})^2 + \mathrm{d}F_i(\xi^{(2)}) + \mathrm{d}^2F_i(\xi^{(1)}, \xi^{(1)}) = 0. \tag{5.4}$$

Here $\mathrm{d}F_i, \mathrm{d}^2F_i$ mean derivatives in the two ordinary coordinates, and parameters are held fixed along the jets. The induced projection on relative logarithmic jets is denoted by $\pi_{[2]}$.

Consider first a field fixing z_1, z_2, z_3, z_4,

$$V = \sum_{i=1}^{2} \sum_{\alpha} v_{i,\alpha} \partial_{a_{i,\alpha}} + \sum_{r=1}^{2} \sum_{j=1}^{4} w_j^{(r)} \partial_{\xi_j^{(r)}}.$$

Write $w^{(r)} = (w_1^{(r)}, w_2^{(r)})$ and associate to its parameter component the polynomial

$$G_{V,i}(y; z, a) = \sum_{\alpha} v_{i,\alpha}(z, a) y^{\alpha}, \qquad y = (y_1, y_2).$$

Differentiating only in the dummy variable y gives the tangency test

$$V(E_{i,0}^{\log}) = G_{V,i}(z), \tag{5.5}$$

$$V(E_{i,1}^{\log}) = \mathrm{d}_y G_{V,i}(z)(\xi^{(1)}) + \mathrm{d}F_i(w^{(1)}) + d_i z_{i+2}^{d_i} w_{i+2}^{(1)}, \tag{5.6}$$

$$\begin{aligned} V(E_{i,2}^{\log}) = {} & \mathrm{d}_y G_{V,i}(z)(\xi^{(2)}) + \mathrm{d}_y^2 G_{V,i}(z)(\xi^{(1)}, \xi^{(1)}) \\ & + \mathrm{d}F_i(w^{(2)}) + 2\mathrm{d}^2F_i(\xi^{(1)}, w^{(1)}) + d_i z_{i+2}^{d_i} w_{i+2}^{(2)} \\ & + 2d_i^2 z_{i+2}^{d_i} \xi_{i+2}^{(1)} w_{i+2}^{(1)}. \end{aligned} \tag{5.7}$$

Tangency means that these expressions vanish on the six incidence equations. For a field with only parameter components it is sufficient that, for both i,

$$G_{V,i}(z) = 0, \qquad \mathrm{d}_y G_{V,i}(z) = 0, \qquad \mathrm{d}_y^2 G_{V,i}(z) = 0. \tag{5.8}$$

These conditions do not involve any jet variable.

5.2. **Third differences and absorption of parameter components.** For $d_i \geqslant 3$, $\alpha \geqslant \gamma$ componentwise, $|\alpha| \leqslant d_i$, and $|\gamma| = 3$, put

$$D_{i,\alpha,\gamma}^{\log} = \sum_{\beta \leqslant \gamma} (-1)^{|\beta|} \binom{\gamma}{\beta} z^{\beta} \partial_{a_{i,\alpha-\beta}}. \tag{5.9}$$

Its parameter polynomial in the i-th block is

$$y^{\alpha-\gamma}(y - z)^{\gamma}.$$

Consequently it annihilates all six equations (5.2)–(5.4), and its coefficients have z-degree at most three.

Lemma 5.1 (Pointwise absorption)**.** *Fix (x, a). The parameter polynomials of $D_{i,\alpha,\gamma}^{\log}$, evaluated at $z = x$, span all polynomials of degree at most d_i whose value, first derivatives, and second derivatives vanish at x. Thus a parameter component satisfying these conditions can be cancelled at (x, a) by a constant linear combination of third differences, without changing any jet component. For $d_i \leqslant 2$, a polynomial with these conditions is already zero.*

Proof. Expand the polynomial in powers $(y-x)^\delta$. Only terms with $3 \leqslant |\delta| \leqslant d_i$ occur. Choose $\gamma \leqslant \delta$ of length three and expand $(y-x)^{\delta-\gamma}$ in powers of y. Each resulting term is $y^\nu(y-x)^\gamma$, with $|\nu+\gamma| \leqslant d_i$, and hence is the parameter polynomial of a field in (5.9). The coefficients of this linear combination are complex numbers depending on the fixed point (x,a). □

We also use the following degree correction. Suppose $d_i \geqslant 2$ and $G^0_{V,i}(y;z,a)$ has y-degree at most d_i+1, with z-degree at most two in its coefficients and with top y-degree part independent of z. Write

$$[G^0_{V,i}]_{d_i+1} = \sum_{|\alpha|=d_i+1} h_{i,\alpha}(a)y^\alpha.$$

For each α fix $\gamma(\alpha) \leqslant \alpha$ of length three. Then

$$G_{V,i} = G^0_{V,i} - \sum_{|\alpha|=d_i+1} h_{i,\alpha}(a)(y-z)^{\gamma(\alpha)}y^{\alpha-\gamma(\alpha)} \tag{5.10}$$

has y-degree at most d_i, coefficient z-degree at most three, and the same value, gradient, and Hessian at $y=z$ as $G^0_{V,i}$. Indeed, the subtraction cancels the top degree and vanishes to order three at $y=z$. Fixing the choices of $\gamma(\alpha)$ makes this construction linear. It is a degree correction before defining the field; pointwise absorption applies afterwards to the resulting degree-d_i parameter polynomial. In degree one we instead impose the compatibility condition below.

5.3. **Matrix fields and quadratic acceleration.** Choose a constant $T \in \mathfrak{sl}_2(\mathbb{C})$, two row vectors $b^{(1)}, b^{(2)} \in (\mathbb{C}^2)^*$, and a symmetric bilinear map $C : \operatorname{Sym}^2\mathbb{C}^2 \to \mathbb{C}^2$. The logarithmic matrix has the same form as in the one-component construction, with one additional normal row:

$$A = \begin{pmatrix} T & 0 & 0 \\ b^{(1)} & 0 & 0 \\ b^{(2)} & 0 & 0 \end{pmatrix} \in M_4(\mathbb{C}).$$

For $d_1, d_2 \geqslant 2$, prescribe a field $V_{A,C}$ fixing the four base coordinates and satisfying

$$\begin{aligned} V_{A,C}(\xi^{(1)}) &= T\xi^{(1)}, \qquad V_{A,C}(\xi^{(2)}) = T\xi^{(2)} + C(\xi^{(1)},\xi^{(1)}), \\ V_{A,C}(\xi^{(1)}_{i+2}) &= b^{(i)}\xi^{(1)}, \quad V_{A,C}(\xi^{(2)}_{i+2}) = b^{(i)}\xi^{(2)}. \end{aligned} \tag{5.11}$$

The matrix field is the case $C=0$; pure quadratic acceleration is the case $A=0$. Before degree correction, the parameter component is represented by

$$\begin{aligned} G^0_{V,i}(y;z,a) = &- \mathrm{d}_yF_i(y,a_i)T(y-z) + d_iF_i(y,a_i)b^{(i)}(y-z) \\ &- \tfrac{1}{2}\mathrm{d}_yF_i(y,a_i)C(y-z,y-z). \end{aligned} \tag{5.12}$$

Its y-degree is at most d_i+1, its coefficients have z-degree at most two, and its top y-degree part is independent of z. The reduction (5.10) gives $G_{V,i}$ of y-degree at most d_i, with coefficients linear in a_i and of z-degree at most three.

For any vectors $r, s \in \mathbb{C}^2$, its Taylor data at $y=z$ are

$$G_{V,i}(z) = 0, \tag{5.13}$$

$$\mathrm{d}_yG_{V,i}(z)(r) = -\mathrm{d}F_i(Tr) + d_iF_ib^{(i)}r, \tag{5.14}$$

$$\begin{aligned} \mathrm{d}^2_yG_{V,i}(z)(r,s) = &-\mathrm{d}^2F_i(r,Ts) - \mathrm{d}^2F_i(s,Tr) - \mathrm{d}F_i(C(r,s)) \\ &+ d_i\big(\mathrm{d}F_i(r)b^{(i)}s + \mathrm{d}F_i(s)b^{(i)}r\big). \end{aligned} \tag{5.15}$$

Here F_i and its derivatives on the right are evaluated at (z,a_i). The factor $1/2$ in (5.12) cancels the two differentiations of the symmetric quadratic expression. Substitution in the

tangency test gives

$$V_{A,C}(E^{\log}_{i,0}) = 0, \tag{5.16}$$

$$V_{A,C}(E^{\log}_{i,1}) = d_i(b^{(i)}\xi^{(1)})E^{\log}_{i,0}, \tag{5.17}$$

$$V_{A,C}(E^{\log}_{i,2}) = d_i(b^{(i)}\xi^{(2)})E^{\log}_{i,0} + 2d_i(b^{(i)}\xi^{(1)})E^{\log}_{i,1}. \tag{5.18}$$

For the cancellation, put $\xi = \xi^{(1)}$ and $\eta = \xi^{(2)}$. Equations (5.6) and (5.14) give

$$\begin{aligned} V_{A,C}(E^{\log}_{i,1}) &= -\mathrm{d}F_i(T\xi) + d_iF_ib^{(i)}\xi + \mathrm{d}F_i(T\xi) + d_iz^{d_i}_{i+2}b^{(i)}\xi \\ &= d_i(b^{(i)}\xi)E^{\log}_{i,0}. \end{aligned}$$

For the second equation, substitution of the gradient and Hessian gives

$$\begin{aligned} V_{A,C}(E^{\log}_{i,2}) &= -\mathrm{d}F_i(T\eta) + d_iF_ib^{(i)}\eta \\ &\quad - 2\mathrm{d}^2F_i(\xi, T\xi) - \mathrm{d}F_i(C(\xi,\xi)) + 2d_i\mathrm{d}F_i(\xi)b^{(i)}\xi \\ &\quad + \mathrm{d}F_i(T\eta) + \mathrm{d}F_i(C(\xi,\xi)) + 2\mathrm{d}^2F_i(\xi, T\xi) \\ &\quad + d_iz^{d_i}_{i+2}b^{(i)}\eta + 2d_i^2z^{d_i}_{i+2}\xi^{(1)}_{i+2}b^{(i)}\xi \\ &= d_i(b^{(i)}\eta)(F_i + z^{d_i}_{i+2}) \\ &\quad + 2d_i(b^{(i)}\xi)\big(\mathrm{d}F_i(\xi) + d_iz^{d_i}_{i+2}\xi^{(1)}_{i+2}\big) \\ &= d_i(b^{(i)}\eta)E^{\log}_{i,0} + 2d_i(b^{(i)}\xi)E^{\log}_{i,1}. \end{aligned}$$

Together with $G_{V,i}(z) = 0$, these identities prove tangency for both components. The degree correction does not alter this calculation, since its value, gradient, and Hessian vanish at $y = z$.

Reparametrization equivariance follows directly from

$$\widetilde{\xi}^{(1)}_j = \lambda\xi^{(1)}_j, \qquad \widetilde{\xi}^{(2)}_j = \lambda^2\xi^{(2)}_j + \mu\xi^{(1)}_j, \qquad j = 1,\dots,4, \quad \lambda \neq 0. \tag{5.19}$$

The linear terms in (5.11) transform linearly and $C(\lambda\xi^{(1)}, \lambda\xi^{(1)}) = \lambda^2C(\xi^{(1)}, \xi^{(1)})$. The parameter coefficients are independent of the jet variables. Thus these fields, and the third differences, commute with the full two-jet reparametrization group.

Lemma 5.2 (Compatibility with a line)**.** *If $d_1 = 1$, the polynomial (5.12) has degree at most one in its first block precisely when*

$$\mathrm{d}F_1(C(r,r)) = 2\mathrm{d}F_1(r)b^{(1)}r \qquad (r \in \mathbb{C}^2). \tag{5.20}$$

In this case

$$G_{V,1}(y; z, a) = \big(-\mathrm{d}F_1T + F_1(z, a_1)b^{(1)}\big)(y - z). \tag{5.21}$$

For $d_2 \geqslant 2$, a finite polynomial family of ten such mixed fields is obtained as follows:

(i) *three basis choices of T, with all other data zero;*
(ii) *two basis choices of $b^{(1)}$, with $C(r,s) = (b^{(1)}r)s + (b^{(1)}s)r$ and $T = b^{(2)} = 0$;*
(iii) *three choices*

$$C(r,r) = \begin{pmatrix} -\partial_2F_1 \\ \partial_1F_1 \end{pmatrix} q(r), \qquad q(r) \in \{r_1^2, r_1r_2, r_2^2\},$$

with all matrix data zero;
(iv) *two basis choices of $b^{(2)}$, with all other data zero.*

After the degree correction in the second block, they satisfy (5.16)–(5.18). At a fixed point where $\mathrm{d}F_1 \neq 0$, their values span all data satisfying (5.20).

Proof. Since $F_1(y) = F_1(z) + \mathrm{d}F_1(y - z)$, the quadratic part of (5.12) vanishes exactly under (5.20); the remaining part is (5.21). The choices in (ii) satisfy this condition directly, and the vector in (iii) is killed by $\mathrm{d}F_1$. For any compatible data, subtract (ii) with the given $b^{(1)}$. The remaining quadratic map takes values in the one-dimensional kernel of $\mathrm{d}F_1$, so it is a linear combination of the three choices in (iii). The T and $b^{(2)}$ components are supplied by (i) and (iv). The normal vector in (iii) depends linearly on the line coefficients and is independent of z. Therefore the second-block correction still has coefficient z-degree at most three. The preceding tangency and equivariance calculations apply to all ten fields. □

5.4. **Projective extension and the common twist.** We first pass from nonzero coefficient vectors a_i to their projective classes. Choose a coefficient linear form ℓ_i; on the open chart $\ell_i(a_i) \neq 0$, represent each class by the vector satisfying $\ell_i(a_i) = 1$. If $v_i = (v_{i,\alpha})$ is a field's coefficient component, it need not preserve this normalization, since $V(\ell_i(a_i)) = \ell_i(v_i)$. Its component tangent to the normalized chart is therefore $v_i - \ell_i(v_i)a_i$.

The auxiliary root must be adjusted at the same time: scaling a_i by a nonzero scalar scales F_i by that scalar, so (5.1) requires the corresponding scaling of z_{i+2} by its d_i-th root. The infinitesimal simultaneous scaling is the radial field in parentheses below. Thus the normalized representative is obtained by

$$V \longmapsto V - \sum_{i=1}^{2} \frac{\ell_i(v_i)}{d_i} \left(d_i \sum_{\alpha} a_{i,\alpha}\partial_{a_{i,\alpha}} + z_{i+2}\partial_{z_{i+2}} \right). \tag{5.22}$$

Indeed, on $\ell_i(a_i) = 1$ the new coefficient component satisfies

$$\ell_i\big(v_i - \ell_i(v_i)a_i\big) = 0.$$

The radial field sends $E^{\log}_{i,\nu}$ to $d_i E^{\log}_{i,\nu}$ and kills the other block's equations, so subtracting its multiple preserves tangency. It also fixes the ordinary base coordinates and every relative logarithmic jet coordinate: its flow multiplies z_{i+2} by a scalar constant along the relative jet, adding only a constant to $\log z_{i+2}$. In particular, if the original field fixes z_{i+2}, its normalized root component is $-\ell_i(v_i)z_{i+2}\partial_{z_{i+2}}/d_i$.

On an overlap of coefficient charts, write $a_i' = s_i(a_i)a_i$. The auxiliary coordinate then changes by $z'_{i+2} = s_i^{1/d_i} z_{i+2}$. Parameters are held fixed along relative jets, so s_i is constant along such jets and $\xi^{(r)}_{i+2}$ is unchanged. In contrast, the vector field may vary the parameters, so $V(s_i)$ need not vanish. If its root component is $\tau_i z_{i+2}\partial_{z_{i+2}}$, the chain rule gives

$$V(z'_{i+2}) = \left(\tau_i + \frac{V(s_i)}{d_i s_i} \right) z'_{i+2}.$$

This is precisely the root correction required when changing the normalized coefficient representative. Thus (5.22) is compatible with changes of coefficient charts. Different choices of the local d_i-th root differ by a locally constant root of unity; the logarithmic jet coordinates and the root field $z_{i+2}\partial_{z_{i+2}}$ are invariant under this change.

We next make the coefficient homogeneities of the generators agree. Here multidegree means the degree of the vector field, including its coordinate derivatives: $\partial_{a_{i,\alpha}}$ has degree -1 in block i. The mixed fields with constant data consequently have multidegree $(0,0)$: their jet coefficients have degree zero, while their i-th coefficient vectors have degree one in a_i, canceled by the degree of $\partial_{a_{i,\alpha}}$. For a line, the kernel accelerations in Lemma 5.2(iii) have multidegree $(1,0)$, because $(-\partial_2 F_1, \partial_1 F_1)^{\mathsf{T}}$ is linear in a_1. An i-th third difference has coefficient components independent of the parameters, hence multidegree -1 in block i and zero in the other block.

Multiplying these three types respectively by the fixed coefficient linear forms

$$\ell_1\ell_2, \qquad \ell_2, \qquad \ell_i^2\ell_j \quad (j \neq i)$$

makes each resulting vector field homogeneous of multidegree $(1,1)$. For example, a third difference in block one has degree $(-1,0)$, and multiplication by $\ell_1^2\ell_2$ raises it to $(1,1)$. After

normalization on each chart by (5.22), this common homogeneity gives sections with parameter twist

$$\mathcal{O}_{\mathbb{P}(V_{d_1})}(1) \boxtimes \mathcal{O}_{\mathbb{P}(V_{d_2})}(1).$$

Indeed, under $a_i' = s_i a_i$, homogeneity contributes the factor $s_1 s_2$, in addition to the coordinate transformation of the vector field; this is the transition rule for the indicated twist. At a fixed point where a multiplier is nonzero, multiplication only rescales the field's value by a nonzero scalar. That scalar can be absorbed into the constants in a pointwise linear combination, so the preceding spanning statements still apply there.

5.4.1. *The base pole bound.* As in the one-component calculation, take the overlap

$$w_0 = 1/z_1, \qquad w_2 = z_2/z_1,$$

and denote its ordinary jets by $\zeta_j^{(r)} = w_j^{(r)}$ for $j = 0, 2$. The inverse jet transformation is

$$\xi_1^{(1)} = -\frac{\zeta_0^{(1)}}{w_0^2}, \qquad \xi_2^{(1)} = \frac{\zeta_2^{(1)}}{w_0} - \frac{w_2\zeta_0^{(1)}}{w_0^2}, \tag{5.23}$$

$$\xi_1^{(2)} = -\frac{\zeta_0^{(2)}}{w_0^2} + \frac{2(\zeta_0^{(1)})^2}{w_0^3}, \qquad \xi_2^{(2)} = \frac{\zeta_2^{(2)}}{w_0} - \frac{w_2\zeta_0^{(2)}}{w_0^2} - \frac{2\zeta_0^{(1)}\zeta_2^{(1)}}{w_0^2} + \frac{2w_2(\zeta_0^{(1)})^2}{w_0^3}. \tag{5.24}$$

For the acceleration part alone these identities give

$$\begin{aligned} V_C(\zeta_0^{(2)}) &= -w_0^2 C^1(\xi^{(1)}, \xi^{(1)}), \\ V_C(\zeta_2^{(2)}) &= w_0\big(C^2(\xi^{(1)}, \xi^{(1)}) - w_2 C^1(\xi^{(1)}, \xi^{(1)})\big). \end{aligned} \tag{5.25}$$

Each $w_0^4 C^j(\xi^{(1)}, \xi^{(1)})$ is polynomial in the new coordinates and first jets. The two displayed components therefore have pole orders at most two and three, respectively. The auxiliary coordinates change to $\widehat{z}_{i+2} = w_0 z_{i+2}$. For the second logarithmic jet, the acceleration contributes

$$V_C\big((\log \widehat{z}_{i+2})''\big) = -w_0 C^1(\xi^{(1)}, \xi^{(1)}), \tag{5.26}$$

also of pole order at most three. The first logarithmic jet is fixed by the acceleration.

If $w = \phi(z)$ and $h = \log w_0$, the matrix contributions to the new first and second ordinary jets are

$$D\phi(T\xi^{(1)}), \qquad D\phi(T\xi^{(2)}) + 2D^2\phi(\xi^{(1)}, T\xi^{(1)}).$$

Those to the new logarithmic jets are

$$\begin{aligned} &b^{(i)}\xi^{(1)} + \mathrm{d}h(T\xi^{(1)}), \\ &b^{(i)}\xi^{(2)} + \mathrm{d}h(T\xi^{(2)}) + 2\mathrm{d}^2h(\xi^{(1)}, T\xi^{(1)}). \end{aligned}$$

Here $D\phi, \mathrm{d}h$ have order at least one in w_0, and $D^2\phi, \mathrm{d}^2h$ have order at least two. Formulas (5.23)–(5.24) show pole order at most two for the terms containing these derivatives; $b^{(i)}\xi^{(2)}$ has order at most three. Interchanging z_1, z_2 handles the other projective overlap.

Every coefficient $v_{i,\alpha}(z, a)$ has z-degree at most three, so

$$w_0^3 v_{i,\alpha}(w_0^{-1}, w_2/w_0, a)$$

is polynomial. The scalar $\ell_i(v_i)$ in (5.22) also has z-degree at most three. Its root component is logarithmic, with coefficient $-\ell_i(v_i)/d_i$, so its pole order is at most three as well. Thus all components extend with base twist $\mathcal{O}_{\mathbb{P}^2}(3)$.

5.4.2. *Regularity along the divisor.* At $C_1 \cap C_2$ the matrix with rows $\mathrm{d}F_1, \mathrm{d}F_2$ is invertible. Equations (5.3) solve for $\xi^{(1)}$ in terms of $\xi_3^{(1)}, \xi_4^{(1)}$ by inverting this regular matrix, and (5.4) then solve for $\xi^{(2)}$. The solutions are regular, linear in the second logarithmic jets and polynomial in the first ones. Substitution into (5.11) proves regularity at the crossing. At a point of only C_i, use (F_i, z_j) as local coordinates and solve for the complementary normal jet using a nonzero partial derivative of F_i. The other auxiliary coordinate is nonzero. These calculations also cover vanishing first jets. All root components are proportional to z_{i+2}, so the auxiliary boundary is preserved. The relation $\mathrm{d}\log F_i = d_i\,\mathrm{d}\log z_{i+2}$ and invariance under the constant root choices prove regular descent to the logarithmic jets downstairs.

Proposition 5.3 (The mixed low-pole fields)**.** *For $d_1 \geqslant 1$, $d_2 \geqslant 2$, the constructions above give a finite family of global logarithmic slanted fields on the relative two-jet space over the SNC parameter locus, with common twist*

$$\mathcal{O}_{\mathbb{P}^2}(3) \boxtimes \mathcal{O}_{\mathbb{P}(V_{d_1})}(1) \boxtimes \mathcal{O}_{\mathbb{P}(V_{d_2})}(1), \tag{5.27}$$

pulled back to that jet space. They fix the base point of $\mathbb{P}^2$ and commute with two-jet reparametrizations.

Proof. Take bases of $T, C, b^{(1)}, b^{(2)}$, giving $3+6+2+2$ mixed fields when both degrees are at least two; use the ten compatible fields of Lemma 5.2 when $d_1 = 1$. Add the finitely many third differences and the indicated parameter multipliers. Tangency, equivariance, and extension have been checked above. □

5.5. **Simultaneous cancellation at a fixed point.** For general parameters a, take a nonempty open set U_a on which $F_1F_2 \neq 0$ and the two rows $\mathrm{d}F_1, \mathrm{d}F_2$ are independent. Fix $x \in U_a$ and a lift $\widetilde{x} = (x, z_3, z_4, a)$ in the chosen coefficient charts. All coefficients in the following construction are evaluated at this point. In the notation of the logarithmic case, put

$$H_i^{\log} = F_i\,\mathrm{d}^2F_i - (\mathrm{d}F_i)^{\mathsf{T}}\mathrm{d}F_i, \qquad i = 1, 2.$$

The value of $G_{V,i}$ at $y = x$ is already zero. Its gradient vanishes precisely when

$$b^{(i)} = \frac{\mathrm{d}F_i(x, a_i)T}{d_iF_i(x, a_i)}. \tag{5.28}$$

Substituting this expression into (5.15) gives

$$F_i\,\mathrm{d}F_i \circ C + T^{\mathsf{T}}H_i^{\log} + H_i^{\log}T = 0, \qquad i = 1, 2, \tag{5.29}$$

as the condition (5.8) for the Hessian of $G_{V,i}$ with respect to y to vanish at $y = x$. Here $(\mathrm{d}F_i \circ C)(r, s) = \mathrm{d}F_i(C(r, s))$, and every term in (5.29) is evaluated at (x, a). The two equations determine a unique symmetric bilinear map C_T:

$$C_T(r, r) = -\begin{pmatrix} F_1\mathrm{d}F_1 \\ F_2\mathrm{d}F_2 \end{pmatrix}^{-1} \begin{pmatrix} r^{\mathsf{T}}(T^{\mathsf{T}}H_1^{\log} + H_1^{\log}T)r \\ r^{\mathsf{T}}(T^{\mathsf{T}}H_2^{\log} + H_2^{\log}T)r \end{pmatrix}. \tag{5.30}$$

The inverse exists by the choice of U_a.

If $d_1 = 1$, then $H_1^{\log} = -(\mathrm{d}F_1)^{\mathsf{T}}\mathrm{d}F_1$. The first equation of (5.29) therefore reads

$$\mathrm{d}F_1(C_T(r, r)) = 2\mathrm{d}F_1(r)\,\frac{\mathrm{d}F_1(Tr)}{F_1} = 2\mathrm{d}F_1(r)b^{(1)}r.$$

This is exactly (5.20). Moreover, (5.21) vanishes identically in y at (x, a), by (5.28). Thus the ten fields of Lemma 5.2 supply the required values of $T, b^{(1)}, b^{(2)}, C_T$ in this case as well.

Lemma 5.4 (Cancellation on a fixed jet fiber)**.** *For each $T \in \mathfrak{sl}_2(\mathbb{C})$ there is a constant linear combination of the fields of Proposition 5.3 whose base, parameter, and auxiliary-coordinate components vanish on the whole jet fiber over $\widetilde{x}$, and whose jet components there are those of (5.11), with $b^{(i)}$ and $C = C_T$ given by (5.28) and (5.30).*

Proof. Choose the mixed fields, or the compatible degree-one fields, with these prescribed values at (x, a). Their parameter polynomials have zero value, gradient, and Hessian at $y = x$. For $d_i \geqslant 3$, cancel the entire polynomial in the i-th block using Lemma 5.1. For $d_i = 2$ it is already zero, and the preceding calculation handles $d_i = 1$. All combinations have constant coefficients once (x, a) is fixed.

This cancellation is compatible with coefficient normalization. Indeed, (5.22) is linear in the parameter components, so cancellation before normalization also cancels the auxiliary radial corrections afterwards. The common-twist multipliers are nonzero scalars at the chosen coefficient point and can be absorbed into the constants in the linear combination. Thus every non-jet component vanishes along the entire fixed fiber. □

Denote the resulting operator on this fixed jet fiber by $\widetilde{V}_T$. In subsequent compositions and commutators, $x, a, T, b^{(i)}, C_T$ are fixed: the operators differentiate only jet variables. In particular, no derivative of the pointwise inverse in (5.30) occurs. Notice that if $T = 0$, the two independent rows $\mathrm{d}F_i$ force $C_T = 0$; obtaining a nonzero acceleration direction requires the commutator calculation below.

5.6. **A commutator at the same fixed point.** At the fixed point, choose the two independent covectors $\mathrm{d}F_1/F_1$ and $\mathrm{d}F_2/F_2$ as a coframe, denoted by $\mathrm{d}z_1, \mathrm{d}z_2$, so that

$$\frac{\mathrm{d}F_1}{F_1} = \mathrm{d}z_1, \qquad \frac{\mathrm{d}F_2}{F_2} = \mathrm{d}z_2.$$

Express both ordinary jet vectors in the dual frame, retaining the notation $\xi_j^{(r)}$ for their components. All matrices and bilinear maps below are expressed in this frame, which is held fixed under the fiber operators. Formula (5.30) becomes

$$C_T^i(r, r) = -\frac{r^{\mathsf{T}}(T^{\mathsf{T}} H_i^{\log} + H_i^{\log} T) r}{F_i^2}, \qquad i = 1, 2.$$

Set

$$W_{12} = \xi_1^{(1)} \xi_2^{(2)} - \xi_2^{(1)} \xi_1^{(2)}.$$

Since $\operatorname{tr} T = 0$, the action on invariant expressions is

$$\widetilde{V}_T(\xi^{(1)}) = T\xi^{(1)}, \qquad \widetilde{V}_T(W_{12}) = \xi_1^{(1)} C_T^2(\xi^{(1)}, \xi^{(1)}) - \xi_2^{(1)} C_T^1(\xi^{(1)}, \xi^{(1)}). \tag{5.31}$$

Use the three matrices

$$\mathsf{H} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad \mathsf{E} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad \mathsf{F} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

With $[V, V'] = VV' - V'V$, their first-jet actions satisfy $[\widetilde{V}_{\mathsf{H}}, \widetilde{V}_{\mathsf{E}}](\xi^{(1)}) = -2\widetilde{V}_{\mathsf{E}}(\xi^{(1)})$. On polynomials in $\xi_1^{(1)}, \xi_2^{(1)}, W_{12}$, the remaining component is

$$\begin{aligned} &[\widetilde{V}_{\mathsf{H}}, \widetilde{V}_{\mathsf{E}}] + 2\widetilde{V}_{\mathsf{E}} \\ &\quad = -2(\xi_2^{(1)})^2 \left[\left(\frac{(H_1^{\log})_{11}}{F_1^2} + \frac{(H_2^{\log})_{12}}{F_2^2} \right) \xi_1^{(1)} + \left(\frac{(H_1^{\log})_{12}}{F_1^2} + \frac{(H_2^{\log})_{22}}{F_2^2} \right) \xi_2^{(1)} \right] \frac{\partial}{\partial W_{12}}. \end{aligned} \tag{5.32}$$

The two cubic expressions entering this calculation are

$$\begin{aligned}\widetilde{V}_{\mathsf{H}}(W_{12}) &= -\frac{2(H_2^{\log})_{11}}{F_2^2}(\xi_1^{(1)})^3 + \frac{2(H_1^{\log})_{11}}{F_1^2}(\xi_1^{(1)})^2\xi_2^{(1)} \\ &\quad + \frac{2(H_2^{\log})_{22}}{F_2^2}\xi_1^{(1)}(\xi_2^{(1)})^2 - \frac{2(H_1^{\log})_{22}}{F_1^2}(\xi_2^{(1)})^3, \\ \widetilde{V}_{\mathsf{E}}(W_{12}) &= -\frac{2(H_2^{\log})_{11}}{F_2^2}(\xi_1^{(1)})^2\xi_2^{(1)} + 2\left(\frac{(H_1^{\log})_{11}}{F_1^2} - \frac{(H_2^{\log})_{12}}{F_2^2}\right)\xi_1^{(1)}(\xi_2^{(1)})^2 \\ &\quad + \frac{2(H_1^{\log})_{12}}{F_1^2}(\xi_2^{(1)})^3.\end{aligned}$$

For this calculation abbreviate

$$u = \xi_1^{(1)}, \qquad v = \xi_2^{(1)}, \qquad h_{i,jk} = \frac{(H_i^{\log})_{jk}}{F_i^2}, \qquad S_{\mathsf{H}} = \widetilde{V}_{\mathsf{H}}(W_{12}), \quad S_{\mathsf{E}} = \widetilde{V}_{\mathsf{E}}(W_{12}).$$

The coefficients $h_{i,jk}$ are constant on the fixed fiber, and $S_{\mathsf{H}}, S_{\mathsf{E}}$ are independent of W_{12}. Hence

$$\begin{aligned}(u\partial_u - v\partial_v)S_{\mathsf{E}} &= -2h_{2,11}u^2v - 2(h_{1,11} - h_{2,12})uv^2 - 6h_{1,12}v^3, \\ v\partial_u S_{\mathsf{H}} &= -6h_{2,11}u^2v + 4h_{1,11}uv^2 + 2h_{2,22}v^3.\end{aligned}$$

Adding twice the displayed expression for S_{E} therefore gives

$$\begin{aligned}\big([\widetilde{V}_{\mathsf{H}}, \widetilde{V}_{\mathsf{E}}] + 2\widetilde{V}_{\mathsf{E}}\big)(W_{12}) &= (-2 + 6 - 4)h_{2,11}u^2v \\ &\quad + \big[-2(h_{1,11} - h_{2,12}) - 4h_{1,11} + 4(h_{1,11} - h_{2,12})\big]uv^2 \\ &\quad + (-6h_{1,12} - 2h_{2,22} + 4h_{1,12})v^3 \\ &= -2v^2\big((h_{1,11} + h_{2,12})u + (h_{1,12} + h_{2,22})v\big).\end{aligned}$$

The operator on the left annihilates u and v, so this proves (5.32) with the stated sign. Only this commutator is needed.

Lemma 5.5 (Generic nonvanishing of the commutator coefficient)**.** *For $d_1 \geqslant 1$ and $d_2 \geqslant 2$, there is a nonempty Zariski-open subset of the SNC parameter space such that, for every parameter a in this subset, there is a nonempty Zariski-open set $U_a \subset \mathbb{P}^2$ on which*

$$F_1F_2\det(\mathrm{d}F_1, \mathrm{d}F_2) \neq 0$$

and the coefficient of $\partial/\partial W_{12}$ in (5.32) is a nonzero polynomial in $\xi_1^{(1)}, \xi_2^{(1)}$ at every point of U_a.

Proof. In the fixed coframe used in (5.32), put

$$A_a(x) := \frac{(H_1^{\log})_{11}}{F_1^2} + \frac{(H_2^{\log})_{12}}{F_2^2}, \qquad B_a(x) := \frac{(H_1^{\log})_{12}}{F_1^2} + \frac{(H_2^{\log})_{22}}{F_2^2}.$$

These are rational functions of the base point and the parameters. The commutator coefficient is

$$-2(\xi_2^{(1)})^2\big(A_a(x)\xi_1^{(1)} + B_a(x)\xi_2^{(1)}\big).$$

At a fixed base point x, this is a nonzero polynomial in the 1-jet variables if and only if

$$(A_a(x), B_a(x)) \neq (0, 0).$$

To verify generic nonvanishing, suppose first that $d_1, d_2 \geqslant 2$ and take, at $x = 0$,

$$F_1 = 1 + z_1 + z_1^2, \qquad F_2 = 1 + z_2 + z_2^2.$$

Here $F_1 = F_2 = 1$, the normalized gradients are the coordinate rows, and

$$H_1^{\log} = \mathrm{diag}(1, 0), \qquad H_2^{\log} = \mathrm{diag}(0, 1).$$

Thus $(A_a(0), B_a(0)) = (1, 1)$. For $d_1 = 1$, take $F_1 = 1 + z_1$ and the same F_2; then $(A_a(0), B_a(0)) = (-1, 1)$. Homogenization realizes these examples in every indicated degree. Their projective zero sets need not be smooth: the examples are used only to show that A_a and B_a do not both vanish identically as rational functions.

After clearing denominators, including those arising from the coframe change, the condition

$$A_a(x) \equiv 0, \qquad B_a(x) \equiv 0$$

is a finite collection of polynomial conditions on the parameters. The examples show that this is a proper closed condition. Its complement therefore meets the dense SNC open subset of the irreducible parameter space. Restrict further so that $F_1F_2 \det(\mathrm{d}F_1, \mathrm{d}F_2)$ is not identically zero on the base.

For each remaining parameter a, let U_a° be the nonempty open subset of the chosen affine chart on which

$$F_1F_2 \det(\mathrm{d}F_1, \mathrm{d}F_2) \neq 0.$$

The functions A_a and B_a are regular there. Define

$$U_a := U_a^\circ \setminus \{x \in U_a^\circ : A_a(x) = B_a(x) = 0\}.$$

Since A_a and B_a do not both vanish identically, U_a is nonempty. At every point of U_a, the commutator coefficient is a nonzero cubic polynomial in the 1-jet variables. □

5.7. **Direct jet differentiation and the second global equation.** We apply Siu differentiation [20, 21, 5] directly to the family ω_1 of Proposition 4.3. Write locally $\omega_1 = Pe$, where e is a frame of $\mathcal{O}_{\mathbb{P}^2}(-t)$, and write a field of Proposition 5.3 as $V = V_\tau \otimes \tau$ in a frame of its common twist. Define $D_V\omega_1 = V_\tau(P)e \otimes \tau$. Under $e' = g(z)e$, one has $P' = g^{-1}P$ and $V_\tau(g) = 0$, because the field fixes the base point; hence $V_\tau(P') = g^{-1}V_\tau(P)$. Under $\tau' = h\tau$, the local field changes to $h^{-1}V_\tau$, which gives the same tensor. Thus the derivatives glue. Let $r_{\lambda,\mu}$ denote the reparametrization action in (5.19). Equivariance gives $r_{\lambda,\mu}^*(V_\tau(P)) = V_\tau(r_{\lambda,\mu}^*P) = \lambda^m V_\tau(P)$, so the weight remains m. With $\mathcal{A} = \mathcal{O}_{\mathcal{S}}(1, 1)|_{\mathcal{U}}$, this yields

$$D_V\omega_1 \in H^0\Big(\mathbb{P}^2 \times \mathcal{U}, E_{2,m}^{\mathrm{rel}} \otimes \big(\mathcal{O}_{\mathbb{P}^2}(-t+3) \boxtimes \mathcal{A}\big)\Big), \tag{5.33}$$

where $E_{2,m}^{\mathrm{rel}}$ is the relative invariant logarithmic jet-differential bundle. The relative direct-image formula (2.3) gives a regular section on the full second Semple space, including Γ_2.

We next show independence by a fixed-point argument. On the complement of the divisor, the local polynomial representing ω_1 has the usual form

$$P(x, a, \xi^{(1)}, \xi^{(2)}) = \sum_{k=0}^{\lfloor m/3 \rfloor} R_k(x, a, \xi^{(1)}) W_{12}^k, \qquad \deg_{\xi^{(1)}} R_k = m - 3k. \tag{5.34}$$

Its pullback by $\pi_{[2]}$ has the same expression; the auxiliary jets are determined by (5.3)–(5.4), since $F_1F_2 \neq 0$. We use the same letter P for this pullback.

For general (x, a), this is a nonzero reduced polynomial in $\xi_1^{(1)}, \xi_2^{(1)}, W_{12}$. Indeed, away from Γ_2 the zero divisor of $\omega_{1,a} = \gamma_a^{b_1} s_a$ is the reduced horizontal divisor Z_a. In characteristic zero its fibers over general base points are reduced. On regular jets, the projection to the Semple space is locally a smooth projection (use a nonzero first-jet coordinate to normalize reparametrizations), so the pulled-back equation is reduced there as well. In the three displayed variables the excluded set $\xi_1^{(1)} = \xi_2^{(1)} = 0$ has codimension two and cannot support a repeated polynomial factor. This proves the assertion after shrinking the open set of base points. No irreducibility of the polynomial at a fixed point is required.

Suppose that every derivative (5.33) vanishes along the relative divisor $\mathcal{Z}$. Choose (x, a) satisfying the preceding reducedness assertion and Lemma 5.5. Each local derivative at this point vanishes on the zero set of the reduced polynomial (5.34); it is therefore divisible by that polynomial. Both have weight m, and all three variable weights $1, 1, 3$ are positive. The

quotient is consequently a scalar. Constant linear combinations and the pointwise cancellation of Lemma 5.4 give

$$\widetilde{V}_T\big(P(x,a,\xi^{(1)},\xi^{(2)})\big) = \rho_{\widetilde{V}_T}(x,a)P(x,a,\xi^{(1)},\xi^{(2)}). \tag{5.35}$$

The numbers $\rho_{\widetilde{V}_T}(x,a)$ are constant under all the fixed-fiber operators. This is the reason for cancelling the parameter components before taking commutators.

Proposition 5.6 (Fixed-point independence)**.** *At a point where the coefficient in* (5.32) *is nonzero, no nonzero polynomial of positive weight of the form* (5.34) *can satisfy* (5.35) *for* $T = \mathsf{H}, \mathsf{E}, \mathsf{F}$.

Proof. The commutator $[\widetilde{V}_{\mathsf{H}}, \widetilde{V}_{\mathsf{E}}]$ annihilates P, since the two proportionality factors in (5.35) are constants on the fixed fiber. Thus (5.32), applied to P, is $2\rho_{\widetilde{V}_{\mathsf{E}}}(x,a)P$. If P has positive degree in W_{12}, its left side is the nonzero coefficient in (5.32) multiplied by

$$\frac{\partial P}{\partial W_{12}} = \sum_{k\geqslant 1} kR_k(x,a,\xi^{(1)})W_{12}^{k-1}.$$

It is nonzero and has smaller W_{12}-degree than P, which is impossible for a scalar multiple of P. Therefore P is independent of W_{12}.

It is now a binary form in $\xi_1^{(1)}, \xi_2^{(1)}$ of degree $m > 0$. The remaining actions of $\widetilde{V}_{\mathsf{E}}$ and $\widetilde{V}_{\mathsf{F}}$ are $\xi_2^{(1)}\partial_{\xi_1^{(1)}}$ and $\xi_1^{(1)}\partial_{\xi_2^{(1)}}$. Both are nilpotent on binary forms of degree m, so their eigenvalues in (5.35) are zero. It follows that both partial derivatives of P vanish. Hence P is constant, contrary to its positive weight. □

The commutator is used only on the fixed fiber to prove independence. The second global equation will be a derivative by one of the original fields, with the twist in (5.27).

Proposition 5.7 ($\mathcal{O}(3)$ slanted derivative)**.** *Consider the family supplied by Proposition 4.3, with its fixed integers* b_1, b_2, m, t, *and assume that* $b_2 > 0$. *After shrinking its nonempty parameter open, write the irreducible reduced horizontal divisor of* $\omega_{1,a}$ *as*

$$Z_a \sim b_1u_1 + b_2u_2 - th, \qquad b_1 \geqslant 2b_2 > 0, \qquad m = b_1 + b_2.$$

There is a further nonempty parameter open $\mathcal{U}$ *and a relative section, with* $\mathcal{A} = \mathcal{O}_{\mathcal{S}}(1,1)|_{\mathcal{U}}$,

$$\Omega_2 \in H^0(\mathcal{X}_2, \mathcal{O}_{\mathcal{X}_2}(m) \otimes \Pi^* \operatorname{pr}_1^* \mathcal{O}_{\mathbb{P}^2}(-t+3) \otimes p_{\mathcal{U}}^*\mathcal{A})$$

whose restriction to Z_a *is nonzero for every* $a \in \mathcal{U}$. *After trivializing* $\mathcal{A}|_a$, *its fiber restrictions correspond to*

$$\omega_{2,a} \in H^0\big(\mathbb{P}^2, E_{2,m}T^*_{\mathbb{P}^2}(\log D_a) \otimes \mathcal{O}_{\mathbb{P}^2}(-t+3)\big)\,.$$

The common zero set of $\omega_{1,a}$ *and* $\omega_{2,a}$ *on* $X_{2,a}$ *has codimension at least two away from* $\Gamma_{2,a}$.

Proof. Intersect the parameter opens in Proposition 4.3 and Lemma 5.5. Each of the finitely many fields in Proposition 5.3 gives the global derivative (5.33) and its corresponding section Ω_V on $\mathcal{X}_2$. If every Ω_V vanished along $\mathcal{Z}$, the fixed-point argument would give (5.35) for all three matrices, contradicting Proposition 5.6. Consequently one of these global fields satisfies

$$\Omega_V|_{\mathcal{Z}} \neq 0. \tag{5.36}$$

The family $\mathcal{Z} \to \mathcal{U}$ is proper. After shrinking $\mathcal{U}$, generic freeness and cohomology and base change identify the spaces of sections of the relevant line bundle on Z_a with the fibers of a vector bundle on $\mathcal{U}$. Equation (5.36) gives a generically nonzero section of that bundle. Removing its proper closed zero locus makes $\Omega_V|_{Z_a}$ nonzero for every remaining parameter. Set $\Omega_2 = \Omega_V$.

Trivializing the parameter factor on each fiber and applying (2.3) yields $\omega_{2,a}$ with the asserted weight and twist. Away from $\Gamma_{2,a}$ the zero divisor of $\omega_{1,a} = \gamma_a^{b_1}s_a$ is the single reduced component Z_a, on which Ω_2 does not vanish identically. Thus the two sections have no common divisorial component there. □

Corollary 5.8. *Let $f : \mathbb{C} \to \mathbb{P}^2$ be algebraically nondegenerate and suppose $t > 3$ and $f^*\omega_{1,a} \equiv 0$. Then either $f^*\omega_{2,a} \not\equiv 0$ for the differential in Proposition 5.7, or the first logarithmic lift $f_{[1]}$ is algebraically degenerate.*

Proof. If both pullbacks vanish, set $Y = \overline{f_{[2]}(\mathbb{C})}^{\mathrm{Zar}} \subset X_2$. By Lemma 2.4, Y is not contained in Γ_2. Its dense open part outside Γ_2 lies in the codimension-two common zero set of the two equations. Since $\dim X_2 = 4$, one has $\dim Y \leqslant 2$. The proper projection $\pi_{2,1}$ therefore maps Y to a proper algebraic subset of the threefold X_1 containing $f_{[1]}(\mathbb{C})$. □

6. The two complementary mechanisms

The argument now follows a fixed decision tree. If the chosen factor does not vanish on the lifted curve, its original twist gives the estimate. If it vanishes, the case $b_2 = 0$ is handled on X_1; for $b_2 > 0$, a pole-three derivative treats $t > 3$, while the key vanishing lemma and the zero-locus theorem treat $t \leqslant 3$.

For one of the six boundary pairs in Lemma 1.2, let $\mathcal{U}_{\mathrm{key}} = \mathcal{U}_{\mathrm{van}}(d_1, d_2)$; for every other degree pair in our range, put $\mathcal{U}_{\mathrm{key}} = \mathcal{S}^{\mathrm{snc}}_{d_1,d_2}$. Let $\mathcal{U}_0$ be the intersection of $\mathcal{U}_{\mathrm{key}}$ with the parameter open furnished by Proposition 4.3. It is nonempty because the parameter space is irreducible. Work first over $a \in \mathcal{U}_0$, and put $D = D_a$ and $\omega_1 = \omega_{1,a}$. Let $f : \mathbb{C} \to \mathbb{P}^2$ be algebraically nondegenerate. If $f^*\omega_1 \not\equiv 0$, Theorem 2.5 and $m/t \leqslant \rho_0^{-1}$ give the required estimate directly. It remains to treat the vanishing case

$$f^*\omega_1 \equiv 0. \tag{6.1}$$

Write the selected horizontal component as

$$Z \sim b_1u_1 + b_2u_2 - th.$$

If $b_2 = 0$, then $Z = \pi_{2,1}^{-1}(Z_1)$ for a proper divisor $Z_1 \subset X_1$. By Lemma 2.4, the regular lift is not contained in Γ_2. Hence (6.1) implies

$$f_{[1]}(\mathbb{C}) \subset Z_1,$$

so $f_{[1]}$ is algebraically degenerate and Theorem 2.6 applies. This branch uses no slanted field. Hence in the two cases below we may assume $b_2 > 0$. Proposition 5.7, applied after restricting the factor family to $\mathcal{U}_0$, supplies a nonempty open subset $\mathcal{U}_1 \subset \mathcal{U}_0$ on which the chosen differential and its slanted derivative are defined simultaneously. In the remainder of this section we work over $a \in \mathcal{U}_1$.

6.1. Case 1: $t > 3$.

Corollary 5.8 supplies an independent

$$\omega_{2,a} \in H^0\Big(\mathbb{P}^2, E_{2,m}\overline{T}^* \otimes \mathcal{O}(-t+3)\Big).$$

If $f^*\omega_{2,a} \not\equiv 0$, Theorem 2.5 gives

$$T_f(r) \leqslant \frac{m}{t-3} N_f^{[1]}(r, D) + o(T_f(r)) \ \|.$$

Since $m/t \leqslant \rho_0^{-1}$ and $t \geqslant 4$,

$$\frac{m}{t-3} \leqslant \frac{1}{\rho_0}\frac{t}{t-3} \leqslant \frac{4}{\rho_0}. \tag{6.2}$$

If $f^*\omega_{2,a} \equiv 0$, Corollary 5.8 makes $f_{[1]}$ algebraically degenerate, and Theorem 2.6 applies.

6.2. **Case 2:** $t \leqslant 3$. Since $b_2 > 0$ and $b_1 \geqslant 2b_2$, one has $m \geqslant 3$. If $\rho_{\mathrm{DEG}} > 1$, then automatically $t/m \leqslant 1 < \rho_{\mathrm{DEG}}$. Otherwise the degree pair is one of the six boundary pairs, and Lemma 1.2 implies that the existing nonzero ω_1 cannot have $t/m \geqslant \rho_{\mathrm{DEG}}$. Thus in every case

$$\rho = \frac{t}{m} < \rho_{\mathrm{DEG}}(d_1, d_2). \tag{6.3}$$

Theorem 2.11 applies. For every rational $0 < \tau < \min\{3, \tau_1(t/m)\}$ it gives either

$$T_f(r) \leqslant \frac{3}{\tau} N_f^{[1]}(r, D) + o(T_f(r)) \ \|$$

or the algebraic degeneracy of $f_{[1]}$, followed by (2.9).

6.3. **The effective constant.** The discrete gap below the zero-locus threshold is explicit. Set

$$\rho_* = \max_{1 \leqslant q \leqslant 3} \frac{q}{\max\{3, \lfloor q/\rho_{\mathrm{DEG}} \rfloor + 1\}}, \qquad \tau_* = \min\{3, \tau_1(\rho_*)\}. \tag{6.4}$$

For integers $1 \leqslant t \leqslant 3$, $m \geqslant 3$ satisfying $t/m < \rho_{\mathrm{DEG}}$, one has $t/m \leqslant \rho_*$. The function $3/\tau_1(\rho)$ is strictly increasing on $[0, \rho_{\mathrm{DEG}})$ by (3.16). Consequently any integer strictly larger than

$$\max\left\{\rho_0^{-1}, \frac{4}{\rho_0}, \frac{3}{\tau_*}, \frac{1}{\kappa}\right\} \tag{6.5}$$

is an admissible $\mathcal{A}_{d_1,d_2}$ in Theorem 1.3. In the order displayed, the four entries account for the original section, its pole-three derivative, the zero-locus section, and McQuillan's branch.

6.4. **The sharper conic–quintic dichotomy.** For $(d_1, d_2) = (2, 5)$ one has

$$\kappa = 4, \qquad \bar{c}_1^2 = 16, \qquad \bar{c}_2 = 21, \qquad \rho_{\mathrm{DEG}} = \frac{19}{48}.$$

The extra case $(m, t) = (8, 3)$ in $\mathfrak{F}_{45}(2, 5)$ changes the useful split from $3/8$ to $1/3$. Put $\rho = t/m$ and assume $b_2 > 0$.

If $\rho \leqslant 1/3$, then $\rho < 19/48$, so Theorem 2.11 applies. Since $3/\tau_1(\rho)$ is increasing on $[0, 19/48)$ by (3.16), it is enough to evaluate the endpoint. Formula (2.13) gives

$$\tau_1(1/3) = \frac{45 - \sqrt{1989}}{6}, \qquad \frac{3}{\tau_1(1/3)} = \frac{45 + \sqrt{1989}}{2} < 45. \tag{6.6}$$

The last inequality follows from $1989 < 45^2$. For the actual ratio $\rho \leqslant 1/3$, one may therefore choose a rational number $1/15 < \tau < \tau_1(\rho)$; the zero-locus coefficient $3/\tau$ is then strictly smaller than 45.

Suppose instead that $\rho > 1/3$. If $t \leqslant 3$, then $m < 3t$ and hence $(m, t) \in \mathfrak{F}_{45}(2, 5)$, contradicting the nonzero choice of ω_1 and Lemma 1.2. Thus $t \geqslant 4$, and the pole-three derivative is available. Integrality gives $m \leqslant 3t - 1$, whence

$$\frac{m}{t-3} \leqslant \frac{3t-1}{t-3} = 3 + \frac{8}{t-3} \leqslant 11. \tag{6.7}$$

If the differentiated equation vanishes as well, the first lift is algebraically degenerate and McQuillan's estimate applies.

Finally,

$$\rho_{\mathrm{RR}}(2, 5) = \frac{2}{9}\left(8 - \sqrt{\frac{455}{8}}\right) > \frac{1}{10}; \tag{6.8}$$

indeed, after squaring, the last inequality is $22750 < 22801$. Choose a rational number

$$\frac{1}{10} < \rho_0 < \rho_{\mathrm{RR}}(2, 5).$$

The direct branch then has coefficient $\rho_0^{-1} < 10$, (6.6) bounds the zero-locus branch by a number strictly smaller than 45, and (6.7) bounds the differentiated branch by 11. McQuillan's coefficient is $1/4$. Therefore the integer 45 covers every branch for a conic and a quintic.

6.5. **The explicit line–octic constant.** For $(d_1, d_2) = (1, 8)$ one has $\kappa = 6$ and $\rho_{\mathrm{DEG}} = 3/8$. Formula (6.4) therefore gives $\rho_* = 1/3$. Substitution in (2.13) yields

$$\tau_1(1/3) = \frac{23 - 5\sqrt{21}}{2}, \qquad \frac{3}{\tau_1(1/3)} = \frac{3}{2}\bigl(23 + 5\sqrt{21}\bigr) < 69. \tag{6.9}$$

Put $\rho = t/m$ and retain the assumption $b_2 > 0$. For $\rho \leqslant 1/3 < \rho_{\mathrm{DEG}}$, the monotonicity on $[0, 3/8)$ proved in (3.16) gives a rational choice $3/69 < \tau < \tau_1(\rho)$ for the zero-locus theorem; hence its coefficient $3/\tau$ is strictly smaller than 69. If $\rho > 1/3$ and $t \leqslant 3$, then $m < 3t$: there are no cases at $t = 1$, and the possibilities are $m = 3, 4, 5$ at $t = 2$ and $m = 3, \ldots, 8$ at $t = 3$. Lemma 1.2 excludes the first three weights at twist 2 and the last three at twist 3; multiplication by a nonzero linear form also excludes the first three at twist 3. Hence, for $\rho > 1/3$, one has $t \geqslant 4$, and integrality gives $m \leqslant 3t - 1$. Thus

$$\frac{m}{t-3} \leqslant \frac{3t-1}{t-3} = 3 + \frac{8}{t-3} \leqslant 11,$$

as in (6.7). If the differentiated equation also vanishes, the first lift is algebraically degenerate and McQuillan's estimate applies. For the direct branch,

$$\rho_{\mathrm{RR}}(1, 8) = \frac{1}{3}\left(8 - \sqrt{\frac{119}{2}}\right) > \frac{2}{23}; \tag{6.10}$$

indeed, after squaring, it reduces to $62951 < 63368$. Choose a rational number

$$\frac{2}{23} < \rho_0 < \rho_{\mathrm{RR}}(1, 8).$$

The direct branch has coefficient $\rho_0^{-1} < 23/2$, the differentiated branch has coefficient at most 11, and the zero-locus coefficient is strictly smaller than 69. McQuillan's coefficient is $1/6$. Thus the integer 69 covers every branch for a line and an octic.

6.6. **The sharper ratio dichotomy for two cubics.** For $(d_1, d_2) = (3, 3)$, the stronger part of Lemma 1.2 improves the preceding general estimate. Put $\rho = t/m$ and retain the assumption $b_2 > 0$.

Suppose first that $\rho \leqslant 1/5$. Since $1/5 < \rho_{\mathrm{DEG}}(3, 3) = 1/4$, Theorem 2.11 applies, irrespective of the absolute size of t. The function $3/\tau_1(\rho)$ is increasing on $[0, 1/4)$ by (3.16), and

$$\tau_1(1/5) = \frac{57 - \sqrt{3189}}{10}, \qquad \frac{3}{\tau_1(1/5)} = \frac{57 + \sqrt{3189}}{2} \approx 56.73561581 < 57. \tag{6.11}$$

Thus one can choose a rational number $3/57 < \tau < \tau_1(\rho)$, and the zero-locus branch has coefficient strictly smaller than 57.

Suppose next that $\rho > 1/5$. If $t \leqslant 3$, integrality gives $m \leqslant 5t - 1$, contradicting the cubic finite vanishing in Lemma 1.2. Hence $t \geqslant 4$, so the mixed $\mathcal{O}(3)$ derivative is available. Since $m \leqslant 5t - 1$,

$$\frac{m}{t-3} \leqslant \frac{5t-1}{t-3} \leqslant 19. \tag{6.12}$$

If the differentiated equation also vanishes on the lifted curve, Corollary 5.8 makes the first lift algebraically degenerate, and Theorem 2.6 applies. Consequently the two branches under $f^*\omega_1 \equiv 0$ and $b_2 > 0$ are bounded by 57; the direct and $b_2 = 0$ branches are included in the final assembly.

7. Proofs of the main statements

We now assemble the preceding alternatives. No new numerical or geometric input is needed at this stage.

Proof of Theorem 1.3. The fixed pair (M,T) of Proposition 4.1 works throughout $\mathcal{S}^{\mathrm{snc}}_{d_1,d_2}$. Intersect the parameter open of Proposition 4.3 with $\mathcal{U}_{\mathrm{key}}$ and call the resulting nonempty open set $\mathcal{U}_0$. The integers b_1, b_2, m, t are fixed on this family. If $b_2 = 0$, put $\mathcal{U}_{\mathrm{SMT}}(d_1,d_2) = \mathcal{U}_0$. If $b_2 > 0$, apply Proposition 5.7 to the restricted family and let $\mathcal{U}_{\mathrm{SMT}}(d_1,d_2) \subset \mathcal{U}_0$ be the nonempty open set it provides.

Fix $a \in \mathcal{U}_{\mathrm{SMT}}(d_1,d_2)$ and choose $\omega_{1,a}$ as in Lemma 4.2 and Proposition 4.3. If $f^*\omega_{1,a} \not\equiv 0$, the jet Second Main Theorem gives the estimate with constant at most ρ_0^{-1}. Under (6.1), the branch $b_2 = 0$ makes $f_{[1]}$ algebraically degenerate and is handled directly by Theorem 2.6. Suppose therefore that $b_2 > 0$. For a degree pair other than $(3,3)$, Case 1 applies when $t > 3$ and gives (6.2) or McQuillan's stronger estimate. When $t \leqslant 3$, Case 2 and (6.4) give the zero-locus estimate or McQuillan's estimate. Hence any integer strictly larger than (6.5) is admissible.

For $(d_1,d_2) = (1,8)$ choose ρ_0 as in (6.10); the preceding line–octic calculation shows that the integer 69 is admissible.

For $(d_1,d_2) = (2,5)$ choose ρ_0 as in (6.8); use the sharper $1/3$ dichotomy of the preceding conic–quintic subsection in place of the coarser general split. The direct, zero-locus, differentiated, and algebraically degenerate branches are then bounded respectively by 10, 45, 11, and $1/4$, with strict inequality where needed. Hence the integer 45 is admissible.

For $(d_1,d_2) = (3,3)$ choose $1/16 < \rho_0 < \rho_{\mathrm{RR}}(3,3)$, so the direct branch has coefficient less than 16. Under (6.1), use instead the sharper ratio dichotomy of the preceding subsection: equation (6.11) gives a coefficient less than 57 when $t/m \leqslant 1/5$, and (6.12) gives a coefficient at most 19 when $t/m > 1/5$. Thus the integer 57 works for two cubics. This proves (1.2). □

Proof of Corollary 1.4. Fix

$$a \in \mathcal{U}_{\mathrm{SMT}}(d_1,d_2) \cap \mathcal{S}_{\mathrm{Chen}}(d_1,d_2)$$

and let

$$f : \mathbb{C} \longrightarrow \mathbb{P}^2 \setminus D_a.$$

If f is algebraically nondegenerate, then $N_f^{[1]}(r, D_a) = 0$, and Theorem 1.3 gives $T_f(r) \leqslant o(T_f(r))$, impossible for a nonconstant curve. If f is algebraically degenerate, Corollary 2.14 makes it constant. Hence the complement is Brody hyperbolic.

It remains to pass from Brody hyperbolicity to the asserted metric statement. The one-dimensional boundary strata are

$$C_1 \setminus C_2, \qquad C_2 \setminus C_1.$$

Since the curves meet transversally in $d_1 d_2$ points, their logarithmic Euler characteristics satisfy

$$2g(C_i) - 2 + d_1 d_2 > 0 \qquad (i = 1,2)$$

throughout the stated degree range. Thus both punctured curves are Brody hyperbolic, and the zero-dimensional stratum $C_1 \cap C_2$ is finite. We use the following form of Green's criterion: if X is projective, $D = \bigcup_{i=1}^{r} D_i$ is a simple-normal-crossing divisor, and both $X \setminus D$ and every open stratum

$$\bigcap_{i \in I} D_i \setminus \bigcup_{j \notin I} D_j \qquad (\varnothing \neq I \subset \{1,\dots,r\})$$

are Brody hyperbolic, then $X \setminus D$ is hyperbolically embedded in X; see [10] and the precise restatement [11, Theorem 4.1]. The strata in the present case are exactly the complement, the two punctured curves above, and the finite set $C_1 \cap C_2$. The criterion therefore shows that the complement is hyperbolically embedded in $\mathbb{P}^2$; in particular it is Kobayashi hyperbolic. The set $\mathcal{U}_{\mathrm{SMT}}(d_1,d_2) \cap \mathcal{S}_{\mathrm{Chen}}(d_1,d_2)$ is very general by the discussion following Corollary 1.4. □

8. Proof of the finite key vanishing lemma: exact finite computation

We now prove Lemma 1.2. It is enough to establish the vanishing at one transverse pair, because upper semicontinuity will then give the required nonempty parameter open. After multiplication by a nonzero homogeneous polynomial reduces to the least twist for each m, the finite range is

$$H^0\big(\mathbb{P}^2, E_{2,m}T^*_{\mathbb{P}^2}(\log D_a)\otimes\mathcal{O}(-t)\big)=0$$

for the pairs listed in the table following Lemma 1.2. The guiding principle is to generate candidate two-jet differentials on one large ordinary chart, test their logarithmic regularity on one mixed logarithmic chart, and impose the available symmetries before constructing any matrix. This chart-and-divisibility strategy starts from the one-component normal form in [7, Proposition 3.1 and Section 3]. The argument below is not a restatement of that calculation: two defining polynomials produce two residue directions, their projective weights must be balanced when $d_1\neq d_2$, and the overlap equations are coupled. For $(3,3)$ we impose the $(\mu_3)^2$ action and coordinate exchange before constructing the matrices. The pairs $(2,e)$, $e=5,6$, have a different cyclic action with coordinate exchange. The line pairs $(1,e)$, $e=8,9$, retain the cyclic action but not the involution, while $(3,4)$ is small enough to run as one unsplit block. The cubic organization is related to the symmetry method developed in [9]. That citation concerns the organizational idea of splitting a finite calculation by symmetry; the two-component normal form, divisibility criterion, matrices, and verification program used here are derived independently in this section. All operations are performed with integral polynomials. Every characteristic-zero conclusion is certified by full column rank at an explicitly recorded good prime; a second prime is used as an independent reproducibility check whenever practical.

The proof has four stages. We first choose explicit SNC test pairs. We then derive a finite ambient normal form and translate regularity on the mixed chart into divisibility equations. Next we decompose these equations by the available finite symmetries. Finally, exact modular row reduction certifies that every resulting block has zero kernel.

8.1. **Degree-uniform test pairs.** We use the following integral pairs; in each row, $A=0$ and $B=0$ are the two components of the test divisor:

(8.1)

(d_1,d_2)	A	B
$(3,3)$	$X^3+2Y^3+Z^3$	$2X^3+Y^3+Z^3$
$(3,4)$	$X^3+Y^3+Z^3$	$X^4+Y^4+Z^4$
$(2,e),\ e=5,6$	$XY+Z^2$	$X^e+Y^e+Z^e$
$(1,8)$	Y	$X^8+Y^8+Z^8+X^4Y^4$
$(1,9)$	Y	$X^9+Y^9+Z^9+X^3Y^3Z^3.$

Each row gives an SNC divisor and remains SNC after adjoining the line $L=\{Z=0\}$. For $(3,4)$, proportional gradients at a common point would force all nonzero coordinates to be equal; neither two nor three equal nonzero coordinates can satisfy the cubic equation. On L, the two equations would give simultaneously $(X/Y)^3=-1$ and $(X/Y)^4=-1$, which is impossible; each curve meets L transversally. The assertions for $(3,3)$ are proved below.

For the rows $(2,e)$, the conic $A=XY+Z^2$ and the Fermat curve $B_e=X^e+Y^e+Z^e$ are smooth. Suppose that they meet nontransversally. If $Z=0$, then $A=0$ forces $XY=0$, which is incompatible with $B_e=0$. Thus we may scale $Z=1$, so $XY=-1$. Writing $\nabla B_e=\lambda\nabla A$, the third component gives $\lambda=e/2$, and the first two components give

$$2X^{e-1}=Y,\qquad 2Y^{e-1}=X.$$

Multiplication by X and Y, respectively, yields $X^e=Y^e=-1/2$. Hence

$$(XY)^e=(-1)^e=X^eY^e=\frac{1}{4},$$

a contradiction for $e=5,6$. On L, the conic meets the line at $[1:0:0]$ and $[0:1:0]$, neither of which lies on B_e; the Fermat curve also meets L transversally. Thus $A+B_e+L$ is SNC.

For $(1,8)$, let

$$B_8 = X^8 + Y^8 + Z^8 + X^4Y^4.$$

At a singular point of B_8, the equation $(B_8)_Z = 8Z^7 = 0$ gives $Z = 0$. Neither X nor Y can vanish, while the other two partial derivatives give

$$2X^4 + Y^4 = 0, \qquad X^4 + 2Y^4 = 0.$$

Their determinant is 3, a contradiction. Thus B_8 is smooth. On the line $A = \{Y = 0\}$ its intersection equation is $X^8 + Z^8 = 0$, so X and Z are nonzero and the intersection is transverse. Smoothness likewise implies transversality with $L = \{Z = 0\}$, because $(B_8)_Z$ vanishes there. Finally, $A \cap L = [1 : 0 : 0] \notin B_8$, and hence $A + B_8 + L$ is SNC.

For $(1,9)$, put

$$B_9 = X^9 + Y^9 + Z^9 + X^3Y^3Z^3.$$

A singular point with a zero coordinate is impossible from the remaining partial derivatives. If all coordinates are nonzero, set $a = X^3$, $b = Y^3$, and $c = Z^3$. The three singularity equations become

$$3a^2 + bc = 3b^2 + ac = 3c^2 + ab = 0.$$

Multiplication gives $27 = -1$, again a contradiction. Hence B_9 is smooth. Its intersections with A and L are transverse by the same restriction argument, and $A \cap L = [1 : 0 : 0] \notin B_9$. Thus $A + B_9 + L$ is SNC.

8.2. **A special symmetric pair of cubic curves.** We work with the following special transverse pair:

$$A = X^3 + 2Y^3 + Z^3, \qquad B = 2X^3 + Y^3 + Z^3. \tag{8.2}$$

Let

$$D = \{AB = 0\}.$$

Both cubics are smooth, and the two curves meet transversally. Indeed, if

$$A = B = 0,$$

then

$$X^3 = Y^3, \qquad 3X^3 + Z^3 = 0.$$

Thus all intersection points have nonzero coordinates. The two gradients cannot be proportional: their Z-components would force the scalar to be 1, whereas the X-components would then require $3X^2 = 6X^2$. Moreover, $A = B = Z = 0$ has no projective solution, and each cubic meets $L = \{Z = 0\}$ transversally. Hence $A + B + L$ is SNC.

The pair has two useful symmetries. First, the involution

$$\sigma(X, Y, Z) = (Y, X, Z) \tag{8.3}$$

satisfies

$$\sigma^* A = B, \qquad \sigma^* B = A.$$

Second, for $\alpha^3 = \beta^3 = 1$, the diagonal transformation

$$[X : Y : Z] \longmapsto [\alpha X : \beta Y : Z] \tag{8.4}$$

fixes both A and B. Thus the finite-dimensional spaces appearing in the computation decompose into characters of $(\mu_3)^2$, and the involution σ exchanges the two character indices.

For the pairs $(2, e)$ in (8.1), the corresponding symmetry is

$$[X : Y : Z] \longmapsto [\zeta X : \zeta^{-1} Y : Z], \qquad \zeta^e = 1,$$

together with the involution $X \leftrightarrow Y$. The monomial

$$x^r y^s (x')^i (y')^j W_{xy}^k$$

has character $r - s + i - j \pmod e$, because W_{xy} has character zero. Coordinate exchange sends the character c to $-c$ and multiplies the k-th Wronskian layer by $(-1)^k$. Consequently the $e = 5$ candidate space splits into six blocks indexed by $\{0\}, \{1,4\}, \{2,3\}$ and the two

involution signs; for $e = 6$ it splits into eight blocks indexed by $\{0\}, \{1, 5\}, \{2, 4\}, \{3\}$ and the two signs.

For either line pair $(1, e)$ the same cyclic transformation

$$[X : Y : Z] \longmapsto [\zeta X : \zeta^{-1} Y : Z], \qquad \zeta^e = 1,$$

preserves both components. There is no coordinate-exchange involution, because the distinguished component is the line $Y = 0$. Since $a = y$ has character -1 on the Z-chart and b is invariant, a source monomial in the k-th Wronskian layer has the shifted character

$$r - s + i - j + n_k \pmod{e}, \qquad n_k = m - 2k. \tag{8.5}$$

Thus the complete candidate space splits into exactly e cyclic character blocks. The shift by n_k is essential: it is contributed by the factor a^{-n_k} in the normal form.

8.3. **The ordinary and mixed logarithmic charts.** The ordinary chart below supports the finite ansatz and makes the finite symmetries transparent. The mixed chart is where we test logarithmic regularity, the negative twist, and the overlaps between the two boundary components.

Let

$$U_0 = \{ZAB \neq 0\}. \tag{8.6}$$

On U_0, put

$$x = \frac{X}{Z}, \qquad y = \frac{Y}{Z},$$

and

$$a = x^3 + 2y^3 + 1, \qquad b = 2x^3 + y^3 + 1. \tag{8.7}$$

The ordinary invariant two-jet frame on U_0 is

$$x', \qquad y', \qquad W_{xy} := x'y'' - x''y'. \tag{8.8}$$

The mixed logarithmic chart is taken in $\{X \neq 0\}$. Put

$$u = \frac{Y}{X}, \qquad z = \frac{Z}{X}.$$

Then

$$a_X = 1 + 2u^3 + z^3, \qquad b_X = 2 + u^3 + z^3. \tag{8.9}$$

Set

$$q = \left(\log \frac{a_X}{b_X} \right)'. \tag{8.10}$$

A direct computation gives

$$q = \frac{3u^2(3 + z^3)}{a_X b_X}\, u' + \frac{3z^2(1 - u^3)}{a_X b_X}\, z'. \tag{8.11}$$

Relative to the ordered ordinary frame (z', u'), the determinant of (z', q) is $\Delta_X/(a_X b_X)$, where the numerator is

$$\Delta_X := a_{X,u} b_X - a_X b_{X,u} = 3u^2(3 + z^3). \tag{8.12}$$

We define

$$U_X = \{X \neq 0,\ u(3 + z^3) \neq 0\}. \tag{8.13}$$

On this chart, the logarithmic invariant two-jet frame is

$$z', \qquad q, \qquad W_X := qz'' - q'z'. \tag{8.14}$$

The two sets U_0 and U_X cover the complement of a finite set. Indeed,

$$\mathbb{P}^2 \setminus (U_0 \cup U_X) \subset \{ZAB = 0\} \cap \{XY(3X^3 + Z^3) = 0\},$$

and the right-hand side is finite because none of the curves $Z = 0$, $A = 0$, or $B = 0$ is a component of $XY(3X^3 + Z^3) = 0$. Since the jet differential sheaves used here are locally free on the normal crossing locus, it is enough to test regularity on $U_0 \cup U_X$.

For the unequal-degree rows of (8.1), the same construction uses the projectively invariant logarithmic form. Put

$$g = \gcd(d_1, d_2), \qquad \alpha = \frac{d_2}{g}, \qquad \beta = \frac{d_1}{g},$$

and on the X-chart define

$$q = \alpha\, d\log a_X - \beta\, d\log b_X, \qquad \Delta = \alpha a_{X,u} b_X - \beta a_X b_{X,u}, \qquad E = \alpha a_{X,z} b_X - \beta a_X b_{X,z}. \tag{8.15}$$

Then

$$q = \frac{\Delta u' + E z'}{a_X b_X}, \qquad u' = \frac{a_X b_X}{\Delta} q - \frac{E}{\Delta} z'. \tag{8.16}$$

The frame generator may replace q by any nonzero rational scalar multiple without changing the kernel problem over $\mathbb{Q}$. The implementation uses this freedom to keep the mixed-chart numerators integral. For the diagonal nonlinear models it uses

$$q_{\mathrm{cmp}} = \frac{1}{d_1} d\log a_X - \frac{1}{d_2} d\log b_X = \frac{g}{d_1 d_2}\, q.$$

For the mixed conic model $a_X = u + z^2$, of bidegree $(2, e)$, it instead uses

$$q_{\mathrm{cmp}} = d\log a_X - \frac{2}{e} d\log b_X = \frac{g}{e} q, \qquad g = \gcd(2, e).$$

Thus $q_{\mathrm{cmp}} = q/5$ for $e = 5$, whereas $q_{\mathrm{cmp}} = q/3$ for $e = 6$. This is the normalization whose degree factors cancel in the corresponding mixed numerators. For the two line models the program uses the integral form q displayed below. Thus in every model the resulting Δ_{cmp}, E_{cmp}, and every matrix entry produced by the program are integers. Each rescaling is an invertible change of frame over $\mathbb{Q}$; it is not being asserted to be invertible after reduction modulo every prime. Instead, the program first constructs its own integral matrix in this normalized frame. Full column rank of that integral matrix modulo one prime implies full column rank over $\mathbb{Q}$, and the rational change of frame then identifies its kernel with the kernel of the invariant form used in (8.15). In particular, the proof-effective prime 5 for the conic–quintic case is legitimate even though 5 divides the denominator in the rational rescaling $q_{\mathrm{cmp}} = q/5$. The order is important: the change of frame is made over characteristic zero, the normalized matrix is then reconstructed with integral entries, and only this integral matrix is reduced modulo 5. No occurrence of the rational scalar $1/5$ is reduced in $\mathbb{F}_5$.

Likewise, the program's cyclic blocks are defined directly by congruences of integer monomial exponents and, where present, by the explicit exchange involution. Their construction does not invoke semisimplicity of a finite-group representation over the proof field. Semisimplicity is used only in characteristic zero to show that these blocks exhaust the candidate space. Hence a proof prime may divide the order of the displayed symmetry group without invalidating the integral block computation. For the two line models these formulas are especially simple. When $e = 8$,

$$\begin{gathered} a_X = u, \qquad b_X = 1 + u^8 + z^8 + u^4, \qquad q = 8\, d\log a_X - d\log b_X, \\ \Delta = 8 + 8z^8 + 4u^4, \qquad E = -8uz^7. \end{gathered} \tag{8.17}$$

When $e = 9$,

$$\begin{gathered} a_X = u, \qquad b_X = 1 + u^9 + z^9 + u^3 z^3, \qquad q = 9\, d\log a_X - d\log b_X, \\ \Delta = 9 + 9z^9 + 6u^3 z^3, \qquad E = -9uz^8 - 3u^4 z^2. \end{gathered} \tag{8.18}$$

In both cases z, u, b_X, Δ are pairwise coprime. For $e = 8$, put $\delta_0 = \Delta/4$. The identity

$$2b_X - \delta_0 = u^4(2u^4 + 1)$$

reduces the only nonobvious gcd to u and the factors of $2u^4+1$; the former does not divide b_X, and no nonconstant polynomial in u divides the polynomial b_X, which is monic in z. For $e=9$, a common factor of b_X and $\Delta = 9b_X - u(b_X)_u$ would, after excluding u, divide $3u^6+z^3$. On each component $z=\lambda u^2$, $\lambda^3=-3$, the restriction of b_X is

$$1-2u^9-27u^{18},$$

which is not identically zero. This proves the asserted coprimality.

The coefficients α,β are exactly what make q independent of the projective trivialization. For every test pair in (8.1), none of Z, A, B is a component of the homogeneous polynomial $X\Delta$. Hence

$$U_0=\{ZAB\neq 0\}, \qquad U_X=\{X\Delta\neq 0\}$$

again cover the complement of a finite set. The general logarithmic frame is (z', q, W_X) with $W_X = qz''-q'z'$. Formulas (8.10)–(8.12) are the specialization $d_1=d_2=3$.

8.4. **An auxiliary logarithmic compactification and a finite ambient normal form.** The ordinary frame on U_0 is convenient for symmetry, but it is not adapted to the line at infinity. In particular, the negative twist cannot be encoded by simply subtracting t from the degree of each numerator in the ordinary frame: cancellations between different ordinary basis terms may remove the apparent poles at infinity. We therefore separate the finite ambient bound from the twisting condition.

Let

$$L:=\{Z=0\}, \qquad D^+:=D+L.$$

For every test pair in (8.1), the divisor D^+ is simple normal crossing. Multiplication by the section $Z^t\in H^0(\mathbb{P}^2,\mathcal{O}_{\mathbb{P}^2}(t))$ gives an injection

$$E_{2,m}T^*_{\mathbb{P}^2}(\log D)\otimes\mathcal{O}_{\mathbb{P}^2}(-t)\xrightarrow{\cdot Z^t} E_{2,m}T^*_{\mathbb{P}^2}(\log D)\hookrightarrow E_{2,m}T^*_{\mathbb{P}^2}(\log D^+). \tag{8.19}$$

The image of the first arrow consists of logarithmic two-jet differentials for D that vanish to order at least t along L. Since $Z=1$ on the affine chart used to define U_0, multiplication by Z^t does not change the local expression on U_0. Thus we may first generate a finite ambient space inside the larger logarithmic bundle for D^+ and impose the removal of the extra logarithmic pole along L, together with the order-t vanishing, on U_X.

We begin with the local pole estimate used for all three components of D^+.

Lemma 8.1 (Layerwise logarithmic pole bound)**.** *Let $s=0$ be a smooth local branch of a logarithmic divisor on a surface, and let v be a transverse coordinate. Put*

$$\lambda=\frac{s'}{s}, \qquad W_{sv}=s'v''-s''v'.$$

Then the logarithmic Wronskian satisfies the triangular identity

$$\lambda v''-\lambda' v'=\frac{1}{s}W_{sv}+\frac{(s')^2v'}{s^2}. \tag{8.20}$$

Consequently, in weighted degree m, the coefficient of the ordinary layer W_{sv}^k has pole order at most

$$n_k:=m-2k$$

along $s=0$. More precisely, the coefficient of

$$(s')^p(v')^{m-3k-p}W_{sv}^k$$

has pole order at most $p+k$.

Proof. Consider a logarithmic monomial

$$\lambda^a(v')^b(\lambda v''-\lambda' v')^r, \qquad a+b=m-3r.$$

A contribution to the ordinary Wronskian layer W^k_{sv} can occur only when $r \geqslant k$. Choose the leading term $s^{-1}W_{sv}$ in exactly k of the r Wronskian factors and the lower term $s^{-2}(s')^2v'$ in the remaining $r-k$ factors. The pole order is then

$$a + k + 2(r-k).$$

The resulting number of factors s' outside W^k_{sv} is

$$p = a + 2(r-k),$$

so the pole order equals $p+k$. Since $p \leqslant m-3k$, this is at most $m-2k$. The statement follows by linearity. □

Write

$$K = \lfloor m/3 \rfloor, \qquad n_k = m-2k, \qquad s_k = m-3k, \qquad d_k^+ = (s-1)m - (2s-1)k.$$

Proposition 8.2 (Finite ambient normal form on U_0)**.** *Every global section of*

$$E_{2,m}T^*_{\mathbb{P}^2}(\log D^+)$$

restricts to U_0 in the form

$$\omega_0 = \sum_{k=0}^{K} \sum_{i+j=s_k} \frac{P_{i,j,k}(x,y)}{a^{n_k}b^{n_k}} (x')^i (y')^j W^k_{xy}, \tag{8.21}$$

where

$$P_{i,j,k} \in \mathbb{C}[x,y], \qquad \deg P_{i,j,k} \leqslant d_k^+ = (s-1)m - (2s-1)k. \tag{8.22}$$

The exponent $n_k = m-2k$ and the degree bound d_k^+ are necessary ambient bounds. Arbitrary polynomials satisfying (8.22) need not define a global section; the missing compatibility conditions are imposed on U_X in the next subsection.

Proof. Apply Lemma 8.1 to the branches $a=0$ and $b=0$. The coefficient of W^k_{xy} has pole order at most n_k along each branch. After multiplication by $a^{n_k}b^{n_k}$, every coefficient is regular on the affine plane, hence is a polynomial. This proves the denominator assertion.

It remains to control the degree by using the extra logarithmic component L. On the X-chart write

$$z = \frac{Z}{X}, \qquad u = \frac{Y}{X}, \qquad x = \frac{1}{z}, \qquad y = \frac{u}{z}.$$

The ordinary jets satisfy

$$x' = -\frac{z'}{z^2}, \qquad y' = \frac{zu' - uz'}{z^2}, \qquad W_{xy} = -\frac{W_{zu}}{z^3}, \tag{8.23}$$

where $W_{zu} = z'u'' - z''u'$. Moreover,

$$a = \frac{a_X}{z^{d_1}}, \qquad b = \frac{b_X}{z^{d_2}}.$$

Thus the k-th layer of (8.21) becomes

$$\frac{z^{(s-2)m-(2s-3)k}}{a_X^{n_k}b_X^{n_k}} \sum_{i+j=s_k} (-1)^{i+k} P_{i,j,k}(z^{-1}, uz^{-1})(z')^i (zu' - uz')^j W^k_{zu}. \tag{8.24}$$

For $0 \leqslant \ell \leqslant s_k$, the coefficient of

$$(z')^{s_k-\ell}(u')^\ell W^k_{zu}$$

is a triangular linear combination of the polynomials $P_{s_k-j,j,k}(z^{-1}, uz^{-1})$ with $j \geqslant \ell$; its diagonal term is a nonzero scalar multiple of

$$z^{(s-2)m-(2s-3)k+\ell} P_{s_k-\ell,\ell,k}(z^{-1}, uz^{-1}).$$

By the refined part of Lemma 8.1, a section logarithmic along $L = \{z=0\}$ may have pole order at most

$$(s_k - \ell) + k$$

in this coefficient. Starting with $\ell = s_k$ and descending in ℓ, the triangularity shows that no polynomial $P_{s_k-\ell,\ell,k}$ can have degree exceeding

$$(s-2)m-(2s-3)k+\ell+(s_k-\ell)+k=(s-1)m-(2s-1)k.$$

This proves (8.22). □

The important point is that the negative twist t does not appear in (8.22). The degree bound describes a finite ambient space with an allowed logarithmic pole along L. The twist is imposed by requiring the transformed coefficients to lose that pole and to vanish to order t along L.

8.5. **Regularity and twisting test on U_X.** On $U_0 \cap U_X$ one has

$$x=\frac{1}{z}, \qquad y=\frac{u}{z}, \qquad a=\frac{a_X}{z^{d_1}}, \qquad b=\frac{b_X}{z^{d_2}}. \tag{8.25}$$

The first derivatives transform as

$$x'=-\frac{z'}{z^2}, \qquad y'=\frac{u'z-uz'}{z^2}. \tag{8.26}$$

Set

$$H:=a_Xb_X, \qquad E:=\alpha a_{X,z}b_X-\beta a_Xb_{X,z}, \qquad \Delta_X:=\alpha a_{X,u}b_X-\beta a_Xb_{X,u}.$$

Equation (8.11) gives

$$u'=\frac{H}{\Delta_X}q-\frac{E}{\Delta_X}z'. \tag{8.27}$$

The pair (z',q) is a logarithmic one-jet frame for D on U_X, and

$$W_X=qz''-q'z'$$

is the associated invariant Wronskian. Substituting (8.25), (8.26), and (8.27) into (8.21) and collecting in the mixed frame gives

$$\omega_0|_{U_X}=\sum_{k=0}^{K}\sum_{j=0}^{s_k}R_{j,k}(u,z)(z')^{s_k-j}q^jW_X^k. \tag{8.28}$$

Proposition 8.3 (Exact extension and twisting criterion)**.** *A candidate* (8.21) *is the restriction of a section*

$$\omega\in H^0\big(\mathbb{P}^2,E_{2,m}T^*_{\mathbb{P}^2}(\log D)\otimes\mathcal{O}_{\mathbb{P}^2}(-t)\big)$$

if and only if every coefficient in (8.28) *satisfies*

$$R_{j,k}\in z^t\mathbb{C}[u,z,\Delta_X^{-1}]. \tag{8.29}$$

Equivalently, after clearing a fixed bounded denominator, the condition is a finite collection of polynomial divisibility equations that are homogeneous and linear in the coefficients of the $P_{i,j,k}$.

Proof. The mixed frame (z',q,W_X) is regular for the logarithmic jet bundle associated with D on U_X. Hence a section of the untwisted bundle is regular on U_X exactly when its scalar coefficients lie in

$$\mathcal{O}(U_X)=\mathbb{C}[u,z,\Delta_X^{-1}].$$

Under the injection (8.19), the image of a section twisted by $\mathcal{O}(-t)$ vanishes to order at least t along $L=\{z=0\}$. This is precisely condition (8.29).

Since $U_0\cup U_X$ misses only finitely many points and the jet bundle is locally free, a section satisfying this condition extends uniquely across the missing codimension-two set. This proves the equivalence. □

For computation, the denominator clearing can be made uniform and explicit. Write

$$u' = Aq + Bz', \qquad A = H/\Delta_X, \qquad B = -E/\Delta_X.$$

Differentiating with respect to the source variable and substituting $u' = Aq + Bz'$ gives the exact identity

$$u'' = Aq' + Bz'' + AA_u q^2 \tag{8.30}$$

$$+ (A_u B + A_z + AB_u)qz' + (BB_u + B_z)(z')^2. \tag{8.31}$$

Consequently,

$$\begin{aligned} W_{zu} = & - AW_X + AA_u q^2 z' \\ & + (A_u B + A_z + AB_u)q(z')^2 + (BB_u + B_z)(z')^3. \end{aligned} \tag{8.32}$$

This is an identity in the localized differential-polynomial ring. Since A, B have one power of Δ_X in the denominator, their first partial derivatives have at most two, and every coefficient in (8.32) has Δ_X-denominator order at most three. A source layer W^r_{zu} can contribute to the target layer W^k_X only when $r \geqslant k$. The $m - 3r$ first-jet factors contribute at most one power of Δ_X^{-1} each, the k leading Wronskian factors contribute one power each, and the remaining $r - k$ lower Wronskian factors contribute at most three powers each. Therefore the total exponent of Δ_X is bounded by

$$(m - 3r) + k + 3(r - k) = m - 2k = n_k. \tag{8.33}$$

No substitution introduces a_X or b_X in a denominator beyond the original source exponent, and $n_r \leqslant n_k$ for $r \geqslant k$.

We also record the z-bound explicitly. Before replacing u' and W_{zu} by the mixed frame, a monomial from source layer r with a numerator of degree d carries the factor

$$z^{sn_r - 2(m-3r) - 3r - d} = z^{(s-2)m - (2s-3)r - d}.$$

Since $d \leqslant (s-1)m - (2s-1)r$, this exponent is at least

$$(s-2)m - (2s-3)r - \big((s-1)m - (2s-1)r\big) = -(m - 2r) = -n_r.$$

The substitutions (8.27) and (8.32) introduce no additional powers of z^{-1}. Hence the pole order in z is at most $n_r \leqslant n_k$. Consequently,

$$\mathcal{N}_{j,k} := z^{n_k} a_X^{n_k} b_X^{n_k} \Delta_X^{n_k} R_{j,k} \in \mathbb{C}[u, z]. \tag{8.34}$$

The same valuation argument applies to every row of (8.1), including the unequal-degree rows. Indeed, $A + B + L$ is SNC, so z, a_X, b_X are pairwise coprime, and the check following (8.16) says precisely that

$$\gcd(z\, a_X b_X, \Delta_X) = 1.$$

Thus localizing at Δ_X changes none of the valuations along the prime factors of $z\, a_X b_X$. This is the required bridge from the general coprimality $\gcd(A, B) = 1$ to the concrete divisibility test.

For example, for the special pair (8.2), the four factors

$$z, \qquad a_X, \qquad b_X, \qquad \Delta_X$$

are pairwise coprime in $\mathbb{C}[u, z]$. Indeed, z divides none of the other three. If an irreducible factor divided both a_X and b_X, it would divide $a_X - b_X = u^3 - 1$; after setting $u^3 = 1$, however, both polynomials become $3 + z^3$, so no factor of $u^3 - 1$ divides them identically. Moreover, u divides neither a_X nor b_X. Finally, for every root ζ of $\zeta^3 = -3$ one has

$$a_X(u, \zeta) = 2(u^3 - 1), \qquad b_X(u, \zeta) = u^3 - 1,$$

which are not the zero polynomial in u; hence no irreducible factor of $3 + z^3$ divides a_X or b_X. Since $\mathbb{C}[u, z]$ is a unique factorization domain, this verifies the general coprimality statement directly in the cubic case. The valuation conditions along $z = 0$, $a_X = 0$, and $b_X = 0$ can

therefore be combined for every test pair. Hence (8.29) is equivalent to the concrete divisibility condition

$$z^{n_k+t}a_X^{n_k}b_X^{n_k} \mid \mathcal{N}_{j,k} \qquad \text{in } \mathbb{C}[u,z]. \tag{8.35}$$

For the conic and cubic models, the fixed product in (8.35) may be handled by the corresponding exact polynomial-remainder test. For a line pair, however, $a_X = u$, and treating the whole product as a univariate polynomial in z would give a nonconstant leading coefficient. We instead use pairwise coprimality and impose the following three equivalent conditions on every cleared numerator $\mathcal{N}_{j,k}$:

(i) no monomial has z-degree below $n_k + t$;
(ii) no monomial has u-degree below n_k;
(iii) the remainder modulo the monic polynomial $b_X^{n_k}$ in $K[u][z]$ is zero, where $K = \mathbb{C}$ for the characteristic-zero system and $K = \mathbb{F}_p$ for a certificate matrix.

The three row families are kept disjoint in the certificate. Hence all divisibility conditions are imposed simultaneously, and each is homogeneous and linear in the coefficients of the $P_{i,j,k}$. The triangularity in the Wronskian degree permits the equations to be generated from the largest k downward.

8.6. **The involution and character reductions.** The involution $\sigma(X, Y, Z) = (Y, X, Z)$ preserves U_0 and acts on the ordinary frame by

$$x \leftrightarrow y, \qquad x' \leftrightarrow y', \qquad W_{xy} \longmapsto -W_{xy}.$$

If one fixes an eigenvalue $\varepsilon = \pm 1$ and imposes

$$\sigma^*\omega_0 = \varepsilon\omega_0, \tag{8.36}$$

then the coefficients in (8.21) satisfy

$$P_{i,j,k}(x, y) = \varepsilon(-1)^k P_{j,i,k}(y, x). \tag{8.37}$$

Thus only one representative from each orbit

$$(i, j, x^r y^s) \longleftrightarrow (j, i, x^s y^r)$$

needs to be generated. If an orbit is fixed and $\varepsilon(-1)^k = -1$, its coefficient is forced to be zero.

The diagonal $(\mu_3)^2$-action gives a further decomposition. For

$$[X : Y : Z] \longmapsto [\alpha X : \beta Y : Z], \qquad \alpha^3 = \beta^3 = 1,$$

one has

$$x \mapsto \alpha x, \qquad y \mapsto \beta y, \qquad x' \mapsto \alpha x', \qquad y' \mapsto \beta y', \qquad W_{xy} \mapsto \alpha\beta W_{xy}.$$

Therefore the monomial

$$x^r y^s (x')^i (y')^j W_{xy}^k$$

has character

$$(r + i + k,\ s + j + k) \pmod 3. \tag{8.38}$$

For a fixed character $(\chi_1, \chi_2) \in (\mathbb{Z}/3\mathbb{Z})^2$, we generate only the monomials satisfying

$$r + i + k \equiv \chi_1 \pmod 3, \qquad s + j + k \equiv \chi_2 \pmod 3. \tag{8.39}$$

The involution σ exchanges the character (χ_1, χ_2) with (χ_2, χ_1). Hence the nine characters split into six orbits:

$$(0,0),\ (1,1),\ (2,2), \quad (0,1) \leftrightarrow (1,0), \quad (0,2) \leftrightarrow (2,0), \quad (1,2) \leftrightarrow (2,1). \tag{8.40}$$

For diagonal characters one imposes σ-eigenconditions inside the character space. For an off-diagonal orbit, an eigenvector is determined by its component in either one of the two exchanged character spaces.

Lemma 8.4 (The cubic symmetry blocks exhaust the candidate space)**.** *Let V be a finite-dimensional candidate space in the above normal form that is stable under both $(\mu_3)^2$ and the involution σ, and let $K \subset V$ be the kernel of the extension and twisting equations. Then $K = 0$ if and only if its intersection with every one of the twelve character–eigenvalue blocks indexed by (8.40) and $\varepsilon \in \{+1, -1\}$ is zero.*

Proof. Over $\mathbb{C}$, the finite abelian group $(\mu_3)^2$ acts semisimply, so $V = \bigoplus_\chi V_\chi$. By Proposition 8.3, the extension and twisting conditions characterize global sections; since the symmetries preserve D and L, their kernel is stable and has the same character decomposition. The involution sends V_χ to $V_{\sigma\chi}$. If $\chi = \sigma\chi$, the involution is diagonalizable with eigenvalues ± 1. If $\chi \neq \sigma\chi$ and $0 \neq v \in K \cap V_\chi$, then at least one of $v + \sigma v$ and $v - \sigma v$ is nonzero; these vectors lie respectively in the $+1$ and -1 eigenspaces of $V_\chi \oplus V_{\sigma\chi}$. Thus every nonzero kernel contains a nonzero vector in one of the listed blocks. The converse is immediate. □

Lemma 8.5 (The conic-pair symmetry blocks exhaust the candidate space)**.** *For either model $(2, e)$, let V be a finite candidate space stable under the cyclic action μ_e and coordinate exchange, and let $K \subset V$ be the kernel of the extension and twisting equations. Then $K = 0$ if and only if its intersection with every character–eigenvalue block indexed by an orbit of $c \mapsto -c$ in $\mathbb{Z}/e\mathbb{Z}$ and by $\varepsilon \in \{+1, -1\}$ is zero.*

Proof. Over $\mathbb{C}$, the cyclic group μ_e acts semisimply. The cyclic action and coordinate exchange preserve D and L, so Proposition 8.3 makes K stable under both and gives $K = \bigoplus_c (K \cap V_c)$. Coordinate exchange sends V_c to V_{-c} and preserves K. On a fixed character ($c = -c$), it is diagonalizable with eigenvalues ± 1. On a two-element orbit $\{c, -c\}$, every nonzero $v \in K \cap V_c$ yields a nonzero vector among $v + \sigma v$ and $v - \sigma v$ in the corresponding $+1$ or -1 block. Thus a nonzero kernel meets one of the enumerated blocks, and the converse is immediate. □

Lemma 8.6 (The line-pair character blocks exhaust the candidate space)**.** *For either line model $(1, e)$, let V be a finite candidate space stable under the cyclic action μ_e, and let $K \subset V$ be the kernel of the extension and twisting equations. Then $K = 0$ if and only if its intersection with each of the e character spaces defined by (8.5) is zero.*

Proof. The cyclic group μ_e acts semisimply in characteristic zero. It preserves D and L, so Proposition 8.3 makes K stable. Both V and K are therefore direct sums of their e character spaces. □

Proposition 8.7 (Finite matrix equivalence)**.** *Fix one of the test pairs in (8.1) and integers $m, t \geqslant 1$. Let V_m be the finite coefficient space in (8.21)–(8.22), and let*

$$\Phi_{m,t} \colon V_m \longrightarrow W_{m,t}$$

record the coefficients of all forbidden remainders in (8.35); for a line pair, $\Phi_{m,t}$ records the three row families listed after that equation. Restriction to U_0 induces a natural isomorphism

$$H^0\big(\mathbb{P}^2, E_{2,m} T^*_{\mathbb{P}^2}(\log D) \otimes \mathcal{O}_{\mathbb{P}^2}(-t)\big) \simeq \ker \Phi_{m,t}.$$

Moreover, the symmetry decompositions above identify this kernel with the direct sum of the kernels of the corresponding block matrices.

Proof. Proposition 8.2 sends every global section to a unique vector of V_m. Proposition 8.3, together with the coprimality and denominator bounds proved above, says that such a vector comes from a twisted global section exactly when all the remainders recorded by $\Phi_{m,t}$ vanish. The resulting section on $U_0 \cup U_X$ extends uniquely across the finite complement. This proves the first assertion. The complete space V_m satisfies the stability hypotheses of the symmetry lemmas: the diagonal actions multiply each normal-form monomial by a scalar, while coordinate exchange permutes the monomials, preserves their total-degree bounds, and preserves the denominator $a^{n_k} b^{n_k}$ whenever that involution is used. Its kernel is also stable, since the symmetries preserve D and L and hence preserve the global extension and twisting conditions. These facts and Lemmas 8.4, 8.5, and 8.6 prove the second. □

The implementation also checks equivariance after the full chart transformation. If a cleared target row contains the monomial $u^U z^Z$, has q-degree Q, and lies in target Wronskian layer k, its character in the source convention is

$$-2U - Z + 2(m - 2k) + Q \pmod{e}. \tag{8.41}$$

For every complete transformed column in the $m = 3$ self-test, each nonzero target row has the same character as the source column. This detects, in particular, omission of the denominator shift in (8.5).

8.7. **Counting the ambient variables.** Let

$$M(d) = \binom{d+2}{2}$$

be the number of monomials in two variables of total degree at most d. Since the negative twist is imposed by the equations on U_X, rather than by deleting monomials on U_0, the ambient variable count depends on m but not on t:

$$N^{+}_{\mathrm{raw}}(m; s) = \sum_{k=0}^{\lfloor m/3 \rfloor} (m - 3k + 1) \binom{(s-1)m - (2s-1)k + 2}{2}. \tag{8.42}$$

For a line pair $(1, e)$, so that $s = e + 1$, this specializes to

$$N_e(m) = \sum_{0 \leqslant k \leqslant \lfloor m/3 \rfloor} (m - 3k + 1) \binom{em - (2e+1)k + 2}{2}. \tag{8.43}$$

The exact counts for the eight line-pair cases are

(d_1, d_2)	(m, t)	$N_e(m)$	blocks	largest block
$(1, 8)$	$(3, 2)$	1,336	8	174
$(1, 8)$	$(4, 2)$	3,077	8	398
$(1, 8)$	$(5, 2)$	6,066	8	779
$(1, 8)$	$(6, 3)$	10,807	8	1,382
$(1, 8)$	$(7, 3)$	17,876	8	2,279
$(1, 8)$	$(8, 3)$	27,849	8	3,543
$(1, 9)$	$(3, 3)$	1,669	9	186
$(1, 9)$	$(4, 3)$	3,857	9	429

(8.44)

Thus the line–octic calculation has 48 blocks and 67,011 ambient columns in total; the line–nonic calculation has 18 blocks and 5,526 columns.

The remainder of this subsection specializes to $(d_1, d_2) = (3, 3)$, so $s = 6$ and the degree bound becomes $5m - 11k$.

For a fixed character (χ_1, χ_2), define

$$M_d(a, b) := \#\{(r, s) \in \mathbb{Z}^2_{\geqslant 0} : r + s \leqslant d,\ r \equiv a \pmod 3,\ s \equiv b \pmod 3\}. \tag{8.45}$$

Then

$$N^{+}_{\chi_1, \chi_2}(m) = \sum_{k=0}^{\lfloor m/3 \rfloor} \sum_{i+j=m-3k} M_{5m-11k}(\chi_1 - i - k, \chi_2 - j - k), \tag{8.46}$$

where all congruences are understood modulo 3. For an off-diagonal character orbit, the number of variables in a σ-eigensystem is the number in either exchanged character. For a diagonal character (χ, χ), the σ-even and σ-odd dimensions are

$$N^{\pm,+}_{\chi}(m) = \frac{1}{2}\left(N^{+}_{\chi,\chi}(m) \pm T^{+}_{\chi}(m)\right), \tag{8.47}$$

where

(8.48)
$$T_\chi^+(m) = \sum_{\substack{0\leqslant k\leqslant \lfloor m/3\rfloor \\ m-3k \text{ even}}} (-1)^k \#\left\{ r \geqslant 0 : 2r \leqslant 5m-11k,\ r+\frac{m-3k}{2}+k \equiv \chi \pmod 3 \right\}.$$

For each twist t, the following table records the endpoint $m = 5t-1$ of the certified cubic range. The last column is the largest block after both the character and involution decompositions.

(8.49)

t	m	N_{raw}^+	$\max N_{\mathrm{sym}}^+$
1	4	1,265	156
2	9	16,511	1,924
3	14	76,905	8,830

Thus the largest ambient candidate space occurs at $(m,t) = (14,3)$ and has 76,905 variables. After the $(\mu_3)^2$-character decomposition and the involution, the largest block has 8,830 variables. The raw ambient total over all pairs

$$1 \leqslant t \leqslant 3, \qquad 1 \leqslant m \leqslant 5t-1$$

is 326,532. The certificate computation instead uses one least twist $t_0(m) = \lceil (m+1)/5 \rceil$ for each $1 \leqslant m \leqslant 14$, including the low-weight self-tests $m = 1, 2$; the raw ambient total for these fourteen cases is 280,798, as recorded in Table 1. The least-twist vanishing covers the larger twists by the injection explained after (8.50). These variables are never placed in one matrix: the computation splits by (m,t), character orbit, and involution eigenvalue. Within each such block, the Wronskian layers organize the downward generation of equations, but the rank test is performed on the full coupled matrix containing all layers. Although the ambient variables do not depend on t, the divisibility equations (8.35) become stronger as t increases.

8.8. **Exact modular rank verification.** For each degree pair and each (m,t) in its finite range, the computation is as follows.

(1) Use the injection (8.19) and generate the finite ambient normal form (8.21) on U_0, with denominator exponent $m-2k$ and degree bound $(s-1)m-(2s-1)k$. Do not subtract t from this degree bound.
(2) For $(3,3)$, split the ambient variables according to the $(\mu_3)^2$-characters and impose the involution eigencondition. For $(2,e)$ use the cyclic character and involution blocks described above. For $(1,e)$ use exactly the e shifted cyclic characters in (8.5), without an involution. Retain one full unsplit sparse basis for $(3,4)$.
(3) Transform the candidate to U_X using (8.25), (8.26), (8.27), and the triangular Wronskian formula (8.32).
(4) Collect the expression in the regular logarithmic frame (z', q, W_X) and form the cleared polynomials $\mathcal{N}_{j,k}$ in (8.34).
(5) Impose the exact divisibility conditions (8.35). For a line pair use the three separate row families listed after that equation. These conditions simultaneously remove the spurious logarithmic pole along L, impose the twist $\mathcal{O}(-t)$, and cancel all forbidden poles along A and B.
(6) Choose a prime for which the adopted normalized integral matrix is defined, reduce that integral matrix to the corresponding finite field, and verify full column rank. If the frame was obtained by a rational rescaling such as $q_{\mathrm{cmp}} = q/5$, perform that rescaling over characteristic zero and reconstruct the integral matrix before reduction; do not reduce the rational change-of-frame scalar itself. Repeat over a second prime as an implementation check; only one proof-effective prime is logically necessary.

For $(3,3)$ it suffices to check the twelve genuine second-level pairs

(8.50)
$$3 \leqslant m \leqslant 14, \qquad t = t_0(m) := \left\lceil \frac{m+1}{5} \right\rceil.$$

Indeed, if $t \geqslant t_0(m)$ and a section with twist $-t$ existed, multiplication by a nonzero form of degree $t - t_0(m)$ would inject it into the space with twist $-t_0(m)$. The implementation also checks $m = 1, 2$ as low-degree self-tests. The remaining minimal pairs are exactly those displayed after Lemma 1.2.

The passage from modular rank to characteristic zero is elementary but decisive. Every cleared matrix has integral entries. Full column rank modulo a prime p exhibits a maximal minor that is nonzero modulo p; that minor is a nonzero integer and remains nonzero over $\mathbb{Q}$.

The certificate program constructs the matrices from (8.35). For every cubic pair in (8.50), all twelve character–eigenvalue matrices have full column rank. Lemma 8.4 then gives zero kernel for the entire cubic candidate space. For $(2, e)$, every cyclic character–eigenvalue block has full column rank; the $(3, 4)$ unsplit matrix does as well. For $(1, e)$, every shifted cyclic character block has full column rank; Lemma 8.6 therefore gives zero kernel. The proof primes are recorded in Table 1. For each block, the computational record gives the degree pair, (m, t), the numbers of rows and columns, the rank, the nullity, the nonzero pivot product, and the prime modulus. The complete computational records and instructions for independent verification are supplied in the supplementary material cited in footnote 1. A separate check confirms that all required cases and blocks are present and that their recorded data are internally consistent. This check is distinct from recomputing the ranks; the supplementary material also provides the means to reproduce the rank computations.

TABLE 1. Finite-rank certificate summary. "Columns" is summed over the indicated independent blocks; in every row the summed rank equals the summed number of columns, so the kernel dimension is zero. The proof modulus is the prime used for the characteristic-zero inference. For $(3, 3)$, the totals include the 12 theorem-relevant pairs in (8.50) and the two low-weight implementation self-tests $m = 1, 2$.

pair	cases	blocks	columns	rank	kernel dim.	proof modulus
$(3, 3)$	14	168	280,798	280,798	0	5
$(3, 4)$	1	1	781	781	0	5
$(2, 6)$	1	8	1,040	1,040	0	7
$(2, 5)$	6	36	38,286	38,286	0	5
$(1, 9)$	2	18	5,526	5,526	0	5
$(1, 8)$	6	48	67,011	67,011	0	5

The proof moduli displayed in the table already suffice. The archive also contains reruns at other primes to test the implementation independently; those reruns are reproducibility checks, not additional mathematical hypotheses. For $(3, 3)$, the newly required case $(m, t) = (9, 2)$ was replayed in full at prime 7. For $(2, 5)$, the additional case $(m, t) = (8, 3)$ has 15,849 columns distributed among six character–eigenvalue blocks. All six blocks have full column rank modulo 5, and an independent replay gives full column rank modulo 7 as well.

Prime 5 is a bad reduction for one $(2, 6)$ block: there the rank is 92/93. This computation serves only as a diagnostic and is not used in the proof. The proof-effective $(2, 6)$ certificate is the full-rank prime-7 run, independently reproduced at prime 11.

For both line pairs, prime 5 supplies the proof-effective full-pivot certificates and prime 11 supplies an independent full-rank replay. Prime 7 is excluded: some relevant integral minors vanish modulo 7, and the degree-nine test curve has bad reduction there because $1^3 = -27$ in $\mathbb{F}_7$. None of the prime-7 diagnostics is used in the proof.

By the maximal-minor argument above, the proof-effective ranks in Table 1 give zero kernel over $\mathbb{Q}$, and hence over $\mathbb{C}$. This proves the required vanishing for every test pair in (8.1).

To pass to general pairs of the fixed degrees, let $\mathcal{U}$ be the Zariski-open parameter space of ordered smooth transverse pairs, let $\mathcal{D} \subset \mathbb{P}^2 \times \mathcal{U}$ be the relative simple-normal-crossing divisor, and let $\mathcal{E}_{m,t}$ be the relative logarithmic invariant two-jet bundle twisted by $\mathcal{O}_{\mathbb{P}^2}(-t)$.

Properness of the projection $\mathbb{P}^2 \times \mathcal{U} \to \mathcal{U}$ and upper semicontinuity of fiberwise cohomology imply that

$$(C_1, C_2) \longmapsto h^0\big(\mathbb{P}^2, E_{2,m}T^*_{\mathbb{P}^2}(\log(C_1 + C_2)) \otimes \mathcal{O}(-t)\big)$$

is upper semicontinuous. Vanishing at the special pair therefore holds on a nonempty Zariski-open subset of $\mathcal{U}$. There are only finitely many pairs (m, t) under consideration; intersecting their nonempty open loci proves Lemma 1.2.

9. The role of the explicit constants

The constants 57, 45, and 69 are explicit choices sufficient to demonstrate the method and its effective Second Main Theorem. More powerful computing equipment and an enlarged certified finite-vanishing range may yield smaller constants, but we do not intend to pursue these improvements. Our aim is to establish the method and the stated degree range; optimizing the numerical coefficients is not essential to that aim.

AI-assisted preparation

Generative artificial-intelligence tools were used for language polishing and, under detailed author direction, to assist in drafting portions of the $\mathcal{O}(3)$ argument and writing the C++ code for the exact finite computations. The authors supplied the core ideas, precise mathematical constraints, and iterative guidance. The ideas and arguments are due to the authors, who checked the C++ code and independently verified every argument in its final form. The authors accept full responsibility for the manuscript and the code.

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